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ORIGINAL ARTICLE

DSLN: Dual-tutor student learning network for multiracial glaucoma detection

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Abstract
Accurate early glaucoma detection is crucial to prevent further vision loss. However, using the off-the-shelf models against fundus image datasets of different races may lead to degraded performance due to domain shift. To address the issue, this paper proposes a dual-tutor student learning network (DSLN) for multiracial glaucoma detection. The proposed DSLN consists of an inter-image tutor, an intra-image tutor, student model and backbone network, which combines the advantages of domain adaptation and semi-supervised learning. The inter-image tutor uses CycleGAN for style transfer to reduce the appearance differences between labeled source domain and labeled target domain images, and transfers the learned knowledge to the student model by minimizing knowledge distillation loss. The intra-image tutor adopts the exponential moving average to leverage the unlabeled target domain and transfers the knowledge to the student model by minimizing prediction consistency loss. Moreover, the student model not only directly learns knowledge from the labeled target domain images, but also learns the intra-image knowledge and inter-image knowledge transfer by two tutors. Furthermore, the backbone integrates the context features of the local optic disc region and global fundus image via modified ResNet50. We conduct extensive experiments on three scenarios constructed from nine public fundus image datasets of three races. Comprehensive experimental results show that the proposed DSLN framework outperforms the state-of-the-art models and has good robustness and generalization: it can effectively overcome domain shift and accurately detect glaucoma from multi-ethnic fundus images.

Keywords Deep learning · Glaucoma detection · Multiracial fundus images · Dual-tutor student model · Domain shift

1 Introduction

Glaucoma is a chronic and irreversible fundus disease, which has become a leading cause of blindness in the world. It has been reported that the number of glaucoma patients will increase to 118 million worldwide by 2040 [1]. The common symptoms of glaucoma are a sudden increase in intraocular pressure, visual field defects and decreased vision. It manifested as large cup-to-disc ratio (CDR), retinal nerve fiber layer defects (RNFLD), peripapillary atrophy (PPA) and hemorrhage in the morphological features of fundus retina [2]. Glaucoma mainly includes open-angle and angle-closure glaucoma [3]. Open-angle glaucoma is the most common, and the illness probability gradually increases with age. There are no obvious clinical symptoms in the early stage because glaucoma is a chronic optic neuropathy, but vision gradually decreases as the disease progresses until it is

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permanently blind. Therefore, early detection and screening play a significant role in preventing vision loss.

Currently, there are three common methods for glaucoma screening: optic nerve head (ONH) assessments, function-based visual field tests and intraocular pressure (IOP) measurement [4]. The ONH assessment is more suitable for large-scale fundus screening, and the CDR is often used for ONH assessment. Figure 1 shows images of healthy eyes and different stages of glaucoma, the purple and blue ellipses represent optic cup (OC) and optic disc (OD), respectively. Cup-to-disk ratio (CDR) represents the ratio between the vertical cup diameter (VCD) and the vertical disc diameter (VDD). The CDR of healthy fundus images is 0.3. The CDR of mild glaucoma is usually between 0.4 and 0.5. The CDR of moderate glaucoma ranges from 0.5 to 0.7, while the CDR of severe glaucoma usually exceeds 0.7.

With the development of imaging technology, 2D fundus photography and 3D optical coherence tomography (OCT) are the most widely used imaging techniques in glaucoma assessment [3]. However, OCT images are expensive and difficult to obtain. Compared with OCT, 2D fundus photography are more widely used in large-scale fundus screening due to its unique advantages of convenience and economy.

Ophthalmologists manually calculate the CDR for glaucoma screening in current clinical applications. However, this method is not only time-consuming and labor-intensive, but also easily affected by subjective factors such as the clinical experience of ophthalmologists, which makes ensuring the accuracy of the diagnosis difficult. Moreover, the phenomenon of domain shift is widespread in various fundus image datasets. The appearance of the fundus images varies due to populations, light source intensities, parameter settings, image resolution and scanner differences. In fact, there are major differences in ONH appearance and feature distribution across different races including differences in CDR, rim size, disc and cup size, which makes it challenging to detect glaucoma accurately. Therefore, it is necessary to create a glaucoma detection method with strong robustness and generalization.

Recently, deep learning has become popular in the medical image analysis field, and numerous convolutional neural network (CNN) methods have been presented to improve the performance of glaucoma detection [5–8]. Currently, two mainstream ideas exist for solving these problems. On the one hand, semi-supervised learning methods are used to exploit abundant unlabeled images to reduce the cost of manual labeling. Wu et al. [5] fully considered the information of undiagnosed fundus images and proposed the teacher-student learning framework for glaucoma classification which combined the optic disc/cup masks and glaucoma labels. They adopted a training strategy of learning to teach with knowledge transfer during training model. Diaz-Pinto et al. [6] applied deep convolutional generative adversarial network to synthesize fundus images and learned feature distribution to classify glaucoma or normal eyes. Although semi-supervised learning methods make full use of the information of unlabeled data, they ignore the feature differences between fundus images of different races. On the other hand, the domain adaptation approaches are used to improve the network generalizability. Wang et al. [7] presented a patch-based Output Space Adversarial Learning (pOSAL) network to jointly segment the OD and OC for glaucoma screening to solve domain shift in diverse retinal image datasets. Liu et al. [8] introduced a collaborative feature-ensembling adaptation (CFEA) framework that used U-Net as a backbone and combined self-ensembling and collaborative adversarial learning to address domain shift in a mutually complementary and beneficial way. These approaches fully use the prior knowledge in labeled images from different datasets, but they ignore the rich unlabeled data features. To the best of our knowledge, there is still a research gap of fully integrating labeled and unlabeled data for accurate glaucoma detection from multi-ethnic fundus images.

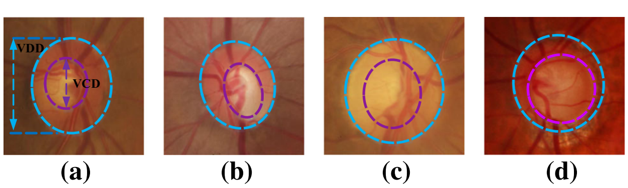

Fig. 1 Normal and different stage of glaucomatous images **a** healthy eyes, **b** early glaucoma, **c** moderate glaucoma and **d** severe glaucoma

Inspired by the above problems, this paper proposes a dual-tutor student learning network (DSLN) that can simultaneously use abundant unlabeled data and limited labeled data for glaucoma detection. We use the teacher-student model as the basic framework and apply two teacher models to train the network. The inter-image tutor utilizes CycleGAN [9] for style transfer from the labeled source images to the labeled target images to reduce the appearance differences, and transfer the learned knowledge to the student model by minimizing the knowledge distillation loss. The intra-image tutor adopts the exponential moving average of the student model to exploit the unlabeled target images, and transfers the prior knowledge to the student model by minimizing the prediction consistency loss. The student model not only directly learns knowledge from the labeled target domain images, but also gets the inter-image and intra-image knowledge transferred by two tutors. We evaluated the proposed DSLN framework under three scenarios constructed from nine public retinal fundus image datasets (e.g., ACRIMA, DRISHTI-

GS and LAG) of three races. The experimental results show that the proposed method has good accuracy and robustness. It can overcome domain shift effectively and detect glaucoma from multi-ethnic fundus images accurately.

The contributions of this paper are as follows:

(1) Considering the huge cost of manually labeling fundus images, this paper proposes a dual-tutor student learning network (DSLN) for automatic glaucoma detection, which fully utilizes the labeled and unlabeled images.
(2) To solve the issue of domain shift between fundus image datasets of different races, this paper employs inter-image tutor and intra-image tutor to learn prior knowledge of labeled source domain images and unlabeled target domain images respectively, and transfer the inter-image knowledge and intra-image knowledge to the student model.
(3) To exploit the rich feature representation in the fundus image, this paper fuses the two streams of the optic disc area and the whole fundus image and adopts modified ResNet50 as the backbone of the tutor and student models.
(4) We perform experiments on three scenes constructed from nine datasets containing 9043 fundus images. Comprehensive experimental results show that the proposed DSLN framework can overcome the domain shift and achieve state-of-the-art detection performance in multi-ethnic fundus images.

The remainder of this paper is organized as follows. Section 2 reviews related works that focus on glaucoma detection. Section 3 introduces the proposed method in detail. Section 4 presents the experimental details and an analysis of the results. Section 5 discusses the advantages and limitations of the proposed method. Section 6 draws conclusions and suggests future work.

2 Related work

Recently the task of glaucoma detection has attracted widespread attention from many researchers. According to the different features used in the detection task, the existing methods are divided into two types: traditional methods and deep learning methods.

2.1 Traditional methods

Traditional methods for glaucoma detection are divided into two categories: non-segmentation-based and segmentation-based methods.

The first non-segmentation-based methods calculate different features such as speeded up robust features (SURF), gray level co-occurrence matrix (GLCM), pyramid histogram of oriented gradients (PHOG), histogram of oriented gradients (HOG) and local binary pattern (LBP) based on traditional image processing methods to perform glaucoma diagnosis. Dutta et al. [10] first used convolution operator and Gaussian filtering to smooth the preprocessed image histogram and estimated the appropriate threshold. Then, they utilized the adaptive threshold based on statistical features to select the threshold from the preprocessed image for optic disc segmentation. Finally, support vector machine (SVM) and neural network classifier are employed to classify glaucoma and normal eyes based on CDR, blood vessel and retinal nerve rim. Tehmina et al. [11] developed a glaucoma diagnosis method based on hybrid texture and structure features, and then used SVM to classify glaucoma and non-glaucoma cases. Joel et al. [12] adopted PHOG and SURF to classify normal and abnormal fundus images automatically. Claro et al. [13] proposed a glaucoma classification method based on morphology, GLCM, HOG, LBP and CNN features.

The second segmentation-based methods for glaucoma detection based on the segmented important regions of the optic disc, optic cup and blood vessels. Kausu et al. [14] first applied Fuzzy C-Means and Otsu's thresholding to segment the optic disc and cup, then extracted CDR and wavelet features and used four classifiers namely random forest, Adaboost, multi-layered perceptrons (MLP) and SVM to classify glaucoma and healthy eyes. Mohamed et al. [15] first used simple linear iterative clustering (SLIC) to segment the input image into superpixels, then applied the statistical pixel level (SPL) to extract features based on histogram of each superpixel. Finally, they utilized a SVM to classify each superpixel into blood vessel, optic disc and optic cup.

In summary, the above traditional methods have achieved good results in automatic glaucoma detection, but they rely on hand-crafted features such as texture, color and shape, and the performance is easily limited by brightness and contrast of fundus images. Therefore, the robustness is poor and not efficient enough to meet the needs of large-scale glaucoma screening.

2.2 Deep learning methods

Recently, the rapid development of deep learning and CNN have resulted in breakthroughs in automatic glaucoma detection methods [16–26].

Current glaucoma detection methods based on deep learning are divided into two types. The first type uses the deep learning network directly to extract the full image features to achieve glaucoma detection. Li et al. [17]

applied Inception-v3 [27] network for training and validation on 48116 color fundus photographs from Chinese to detect glaucoma. The AUC value, sensitivity and specificity of this method on the validation set reached 98.6%, 95.6% and 92%, respectively. The study found that high or pathological myopia would produce false negative results, while physiologic large cupping and pathologic myopia would cause false positive results. Liu et al. [18] adopted the ResNet [28] to train and validate on 269,601 fundus images of Chinese. The AUC value, specificity and sensitivity of the model on the validation set reached 99.6%, 97.7% and 96.2%, respectively. Christopher et al. [19] used three CNN models of VGG16 [29], Inception-v3 [27] and ResNet50 [28] to evaluate 14822 fundus photographs from different races and countries for glaucoma detection. Two learning strategies of native and transfer learning are used to train each network. Experimental results showed that compared with other network, the combination of ResNet and transfer learning is better for glaucoma diagnosis. Wang et al. [30] proposed a multichannel-based generative adversarial network (MGAN) with semisupervision for grading DR, which makes full use of both unlabeled data and labeled data. Yu et al. [31] proposed a tensorizing GAN with high-order pooling, which combined tensor-train decomposition, higher-order pooling and semisupervised learning, for Alzheimer's disease and mild cognitive impairment diagnosis.

The second type uses deep learning network to segment the optic disc, optic cup, blood vessel and other related tissues in fundus images, and then calculate clinical measures for glaucoma classification. Inspired by the issue that most misclassified glaucoma samples focus on high redundancy in fundus images, Li et al. [20] introduced attention mechanism to provide the network with prior knowledge from ophthalmologists to reduce the redundancy for accurate glaucoma detection. They proposed a three-stage CNN framework which including three parts: attention prediction, pathological area localization and glaucoma classification. In addition, they collected 5824 fundus images from Beijing Tongren Hospital to build a large-scale attention-based glaucoma (LAG) dataset with corresponding attention map. Fu et al. [21] proposed a disc-aware ensemble network (DENet), which fused the features of the local optic disc area and the whole fundus image for automatic glaucoma detection. They conducted experiments on 10109 fundus images from Singapore and obtained good results. Bisneto et al. [22] combined cGAN, taxonomic diversity and discrimination index to distinguish between normal eyes and glaucoma in the Drishti-GS [32] and RIM-ONE [33] datasets. Chai et al. [23] developed a multi-branch neural network that fused image features (e.g., CDR, PPA and RNFLD) and non-image features (e.g., age, eyesight and intraocular pressure), and performed experiment on 2554 images for automatic glaucoma diagnosis. Metha et al. [24] adopted InceptionResnet-v4 [34] architecture to experiment on rich clinical data, color fundus images and OCT from UK Biobank for glaucoma screening. The study showed that fusion of multi-source information is beneficial to improve the performance of glaucoma detection. Bajwa et al. [25] proposed a two-stage approach for optic disc localization and glaucoma classification on HRF [35], ORIGA [36] and OCT & CFI [37] datasets. This method first used Faster R-CNN to locate and extract the optic disc from the fundus images, and then adopted a four-layer CNN to classify healthy eyes and glaucoma. Instead of using CDR, Kumar et al. [26] used rim-to-disc ratio (RDR) and disc-damage-likelihood scale to perform glaucoma prescreening and severity assessment on 3323 fundus images from public and private datasets.

Most of the above methods have achieved better performance on fundus images of a certain race or country, but the robustness of existing methods on fundus images of multiple countries and races needs to be improved.

The goal of this study is to create a glaucoma detection method that overcomes the limitations of traditional and deep learning methods. Therefore, in this paper, we proposed a dual-tutor student learning network (DSLN) for accurate glaucoma detection from multi-ethnic fundus images.

3 Proposed methods

3.1 Problem formulation

Considering a labeled source dataset $S = \{(x_i^s, y_i^s)\}_{i=1}^{m_s}$ and a labeled target dataset $T = \{(x_j^t, y_j^t)\}_{j=1}^{m_t}$, where x_i^s represents the $i-th$ image in the labeled source domain, y_i^s represents the image-level label corresponding to the source domain image x_i^s, and m_s represents the number of labeled source domain images. Similarly, x_j^t represents the $j-th$ image in the labeled target domain, y_j^t represents the image-level label corresponding to the target domain image x_j^t, and m_t denotes the number of images. Moreover, given an unlabeled target dataset $U = \{x_k^u\}_{u=1}^{m_u}$, where x_k^u represents the $k-th$ image of the unlabeled target domain, and m_u represents the number of images. Assume that S and T come from fundus image dataset of two races, and m_u are much larger than m_t. Our goal is to make full use of the prior knowledge in S, T and U for improving the classification performance on the target domain images.

3.2 Framework overview

The overall framework of the proposed DSLN is shown in Fig. 2. The network is composed of four parts: inter-image tutor model, intra-image tutor model, student model and backbone network. The inter-image tutor and intra-image tutor models learn the prior knowledge from the labeled source domain images and unlabeled target domain images, respectively, and transfer the acquired knowledge to the student model at the same time for comprehensive integration and optimization.

3.3 Inter-image tutor

Although the source domain and target domain images have some similar prior knowledge, they are different in appearance and feature distribution. Therefore, we expect to reduce the appearance difference between different domain images for appearance alignment. First, we use CycleGAN [9] to transfer style from the source images x^s to the synthetic target images $x^{s\to t}$ and obtain the synthetic target domain $D^{s\to t}$. The realization process is shown in Fig. 3.

CycleGAN [9] includes two pairs of generator and discriminator networks: $G_{s\to t}, D_t, G_{t\to s}, D_s$. The generator adopts the encoder-decoder framework as the backbone, and the transformer consists of eight residual blocks. The goal of the target domain generator $G_{s\to t}$ is to synthesize images with target domain style to confuse the discriminator, while the target domain discriminator D_t is to distinguish between the fake generated images $x^{s\to t}$ and real target domain images x^t. Similarly, the source domain generator $G_{t\to s}$ can transform the images with style transfer into the original images, while the discriminator D_s can distinguish between the fake generated images $x^{t\to s}$ and real source domain images x^s. As shown in Fig. 3, the paired generators and discriminators are represented by the same color, and the green and yellow lines represent the data flow of the target domain image and the source domain image. The paired generators and discriminators act a two-player minimax game and follow the adversarial loss for optimization:

$$\begin{aligned} &L_{GAN} = L_{GAN}^S + L_{GAN}^T, \\ &L_{GAN}^S = E_{x^s\sim S}[logD_s(x^s)] + E_{x^t\sim T}[log(1 - D_s(G_{t\to s}(x^t)))], \\ &L_{GAN}^T = E_{x^t\sim T}[logD_t(x^t)] + E_{x^s\sim S}[log(1 - D_t(G_{s\to t}(x^s)))]. \end{aligned} \tag{1}$$

Furthermore, we introduce cycle consistency constrain L_{CYC} to encourage $G_{s\to t}(G_{t\to s}(x^t)) \approx x^t$ and $G_{t\to s}(G_{s\to t}(x^s)) \approx x^s$ in the transformation:

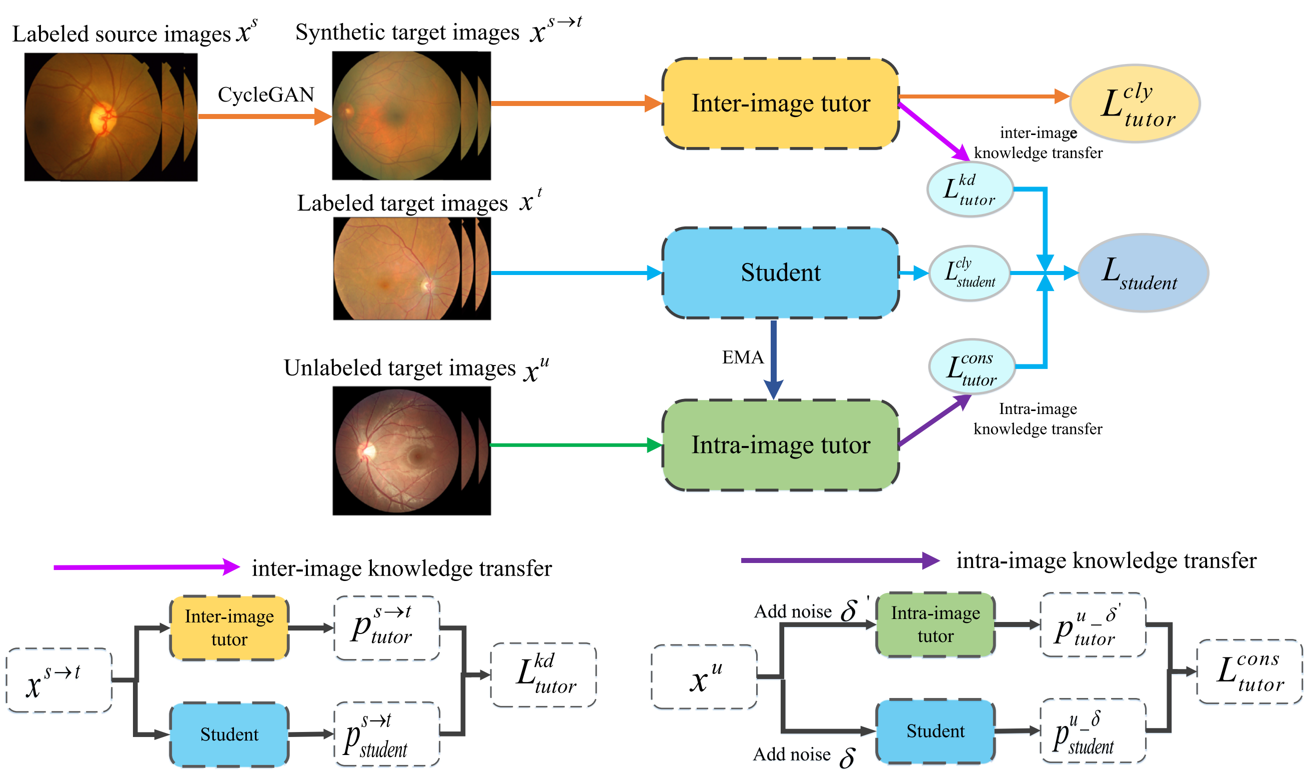

Fig. 2 Overall architecture of proposed DSLN framework

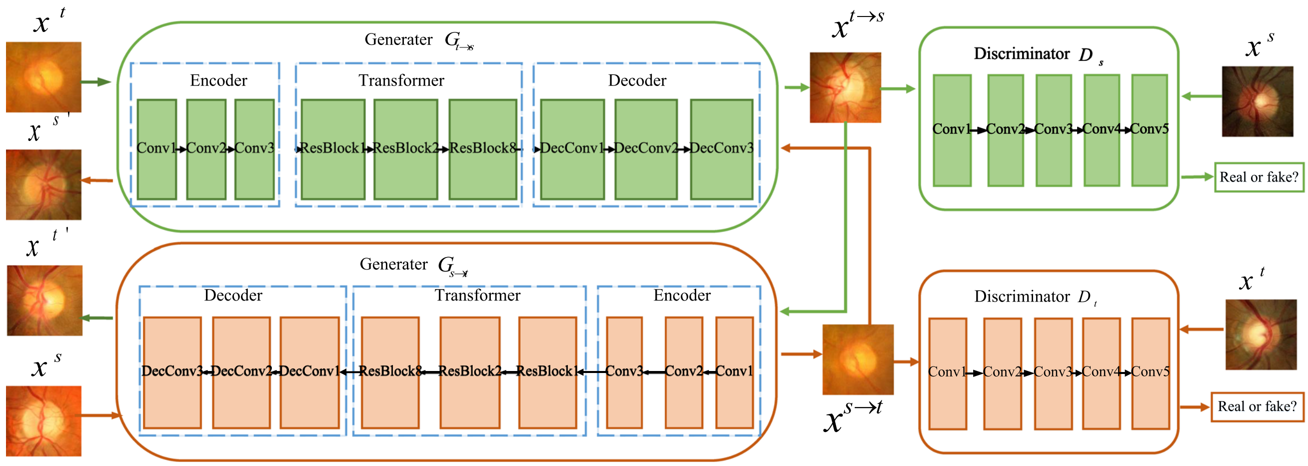

Fig. 3 Implementation flow of CycleGAN

$$L_{CYC}(G_{s\to t}, G_{t\to s}) = E_{x^s\to S}[\|G_{t\to s}(G_{s\to t}(x^s)) - x^s\|_1] + E_{x^t\to T}[\|G_{s\to t}(G_{t\to s}(x^t)) - x^t\|_1]. \quad (2)$$

Therefore, the overall objective function of CycleGAN can be expressed as follows:

$$L_{CycleGAN} = L_{GAN} + \omega L_{CYC}, \quad (3)$$

where ω is a factor used to control the relative importance of the two objectives.

Then, we input the synthesized target domain images $x^{s\to t}$ and labels y^s into the inter-image tutor to learn knowledge from the source domain. The learning function is defined as follows:

$$\begin{aligned} L_{\text{tutor}}^{cly} &= \alpha_1 L_{\text{Dice}}\left(y^s, p_{\text{tutor}}^{s\to t}\right) + \alpha_2 L_{\text{BCE}}\left(y^s, p_{\text{tutor}}^{s\to t}\right) \\ &= \alpha_1\left(1 - \frac{2\langle y^s, p_{tutor}^{s\to t}\rangle}{\|y^s\|_1 + \|p_{\text{tutor}}^{s\to t}\|_1}\right) \\ &\quad + \alpha_2\left[-\kappa y^s log p_{tutor}^{s\to t}\right. \\ &\quad \left. -(1-\kappa)(1-y^s)log(1-p_{tutor}^{s\to t})\right], \end{aligned} \quad (4)$$

where L_{Dice} and L_{BCE} represent dice loss and class-balanced binary cross-entropy loss, respectively. $p_{\text{tutor}}^{s\to t}$ represent the prediction result of inter-image tutor with $x^{s\to t}$ as input, α_1 and α_2 are balance factors, which are used to balance the weights of L_{Dice} and L_{BCE}, respectively. $\langle y^s, p_{tutor}^{s\to t}\rangle$ denotes the matrix dot product of source domain labels and prediction results of inter-image tutor, and $\|\cdot\|_1$ denotes a norm, i.e., the sum of the absolute values of the matrix elements. κ and $1-\kappa$ indicate the weighting factor used to balance the difference between glaucoma and normal pixels.

Finally, to transfer the learned knowledge from the inter-image tutor model to the student model, we input the synthesized target domain image $x^{s\to t}$ into the student and inter-image model, respectively, and expect the student model produce prediction results similar to the inter-image tutor by minimizing the knowledge distillation loss L_{tutor}^{kd}. We define L_{tutor}^{kd} as follows:

$$\begin{aligned} L_{\text{tutor}}^{kd} &= L_{\text{BCE}}\left(p_{\text{student}}^{s\to t}, p_{\text{tutor}}^{s\to t}\right) \\ &= -\left[\kappa p_{\text{student}}^{s\to t} log p_{tutor}^{s\to t}\right. \\ &\quad \left. +(1-\kappa)(1-p_{\text{tutor}}^{s\to t}) log p_{\text{student}}^{s\to t}\right], \end{aligned} \quad (5)$$

where $p_{\text{student}}^{s\to t}$ and $p_{\text{tutor}}^{s\to t}$ represent the prediction results of the student model and the inter-image tutor model, respectively. κ and $1-\kappa$ indicate the weighting factor used to balance the difference between glaucoma and normal pixels.

3.4 Intra-image tutor

Since there is no image-level label in U to supervise network learning while there is manual label in T, therefore, we imitate the student model framework to construct the intra-image tutor model to make full use of the rich feature representation in U and T. The intra-image tutor weight λ' is updated according to the exponential moving average (EMA) of student model weight λ in the training process. Specifically, the update process of intra-image tutor model weight can be expressed as follows:

$$\lambda' = \beta\lambda'_{c-1} + (1-\beta)\lambda_c, \quad (6)$$

where c represents training step, and β represents EMA delay rate used to control update. To transfer the learned knowledge from the intra-image tutor to the student, we add different random Gaussian noise δ and δ' to the same unlabeled target images and input them into the student model and the intra-tutor model, respectively. Considering that Gaussian noises cannot cause strong perturbation to the images, we expect the student model and the intra-image tutor model to output similar predictions. Therefore, we encourage the student model and the intra-image tutor model to produce consistent predictions by minimizing the consistency loss. The consistency loss L_{tutor}^{cons} is defined as follows:

$$\begin{aligned} L_{\text{tutor}}^{\text{cons}} &= L_{MSE}\left[p_{\text{intra}}(x^u;\delta^{'};\lambda^{'}), p_{\text{student}}(x^u;\delta;\lambda)\right] \\ &= \frac{1}{m_u}\sum_{u=1}^{m_u}\left[p_{intra}(x^u;\delta^{'};\lambda^{'}) - p_{\text{student}}(x^u;\delta;\lambda)\right]^2, \end{aligned} \tag{7}$$

where L_{MSE} represents the mean square error loss. $p_{\text{intra}}(x^u;\delta^{'};\lambda^{'})$ and $p_{\text{student}}(x^u;\delta;\lambda)$ represent the predictions of the intra-image tutor model with noise $\delta^{'}$ and weight $\lambda^{'}$ and the student model with noise δ and weight λ, respectively.

3.5 Student model

For the student model, its learning function under the supervision of the labeled target domain image and the corresponding label is defined as:

$$\begin{aligned} L_{\text{student}}^{cly} &= \eta_1 L_{Dice}(y^t, p_{\text{student}}^t) + \eta_2 L_{BCE}(y^t, p_{\text{student}}^t) \\ &= \alpha_1\left(1 - \frac{2\langle y^t, p_{\text{student}}^t\rangle}{\|y^t\|_1 + \|p_{\text{student}}^t\|_1}\right) \\ &\quad + \alpha_2\left[-\kappa y^t log p_{\text{student}}^t \right. \\ &\quad \left. -(1-\kappa)(1-y^t)log(1-p_{\text{student}}^t)\right], \end{aligned} \tag{8}$$

where p_{student}^t represents the prediction result of student model on the labeled target domain T. η_1 and η_2 are hyperparameters used to balance the weights of L_{Dice} and L_{BCE}, respectively. $\langle y^t, p_{\text{student}}^t\rangle$ denotes the matrix dot product of target domain labels and prediction results of student model.

The student model not only learns the prior knowledge of T, but also learns the knowledge of S and U. Specifically, the source of learning knowledge of the student model includes three parts: knowledge distillation loss L_{tutor}^{kd} to obtain inter-image knowledge, prediction consistency loss L_{tutor}^{cly} to obtain intra-image knowledge, and knowledge L_{student}^{cly} from labeled target domains. In summary, the objective function of the student model can be expressed as:

$$L_{\text{student}} = \gamma_1 L_{\text{tutor}}^{kd} + \gamma_2 L_{\text{tutor}}^{\text{cons}} + L_{\text{student}}^{\text{cly}}, \tag{9}$$

where γ_1 and γ_2 represent the factors used to balance the weight of L_{tutor}^{kd} and L_{tutor}^{cons}, respectively.

Our proposed method updates the training network in an end-to-end manner. The input is the original fundus images, and the output is the binary classification results of glaucoma and normal images. First, we learn knowledge from source domain and optimize the inter-image tutor model through appearance alignment, and then use the EMA of the student model to update the intra-image tutor model. Finally, the inter-image knowledge, intra-image knowledge and the knowledge learned from labeled target images are integrated to optimize the student model. In this way, the tutor model and the student model can update the learning parameters synchronously in an online manner.

3.6 Backbone network

The backbone network of our proposed DSLN framework integrating the features extracted from whole fundus image and local optic disc region for glaucoma detection. The global fundus image provides the general structural features, while the optic disc area contains high-level detailed information. The two streams provide guidance for glaucoma detection tasks from different aspects.

First, we use the two-stage object detection framework Faster R-CNN to locate the optic disc region. As shown in Fig. 4, Faster R-CNN mainly includes the following three major components: (1) a feature extraction network, (2) a region proposal network (RPN) that simultaneously predicts object boundaries and improves object scores, and (3) regions of interest (RoI) header, including bounding box regression and classifier for post-processing the regression and classification results. In the optic disc region location module, we adopt VGG16 to extract the feature map of the original fundus image, and then used RPN to generate region proposals. RPN determines whether anchors are positive or negative according to softmax and utilizes bounding box regression to correct anchors to get accurate proposals. The ROI pooling layer extracts proposal feature maps and sends them to the subsequent fully connected layer for object classification. The classifier in ROI header calculates the proposal category, meanwhile the bounding box regression gets the final accurate location of the optic disc. In Fig. 4, the cyan rectangle represents the convolution operation, the gold represents the max-pooling, the green represents the region proposal network, the orange represents the ROI pooling layer, the purple represents the fully connected layer, and the blue represents the bounding box regression and classifier in the ROI header.

Second, the detected optic disc region is resized to 224×224 and input to the modified ResNet50. The modified ResNet50 consists of five convolution blocks, followed by a global max pooling layer, a detailed convolution layer and an up-sampling layer. Different from the original ResNet50, we use dilated convolution with dilation rate of 2 in the last two convolution blocks instead of the standard convolution to extract the features of different receptive fields. In addition, we employ four parallel 3×3 dilated convolution branches to replace the last fully

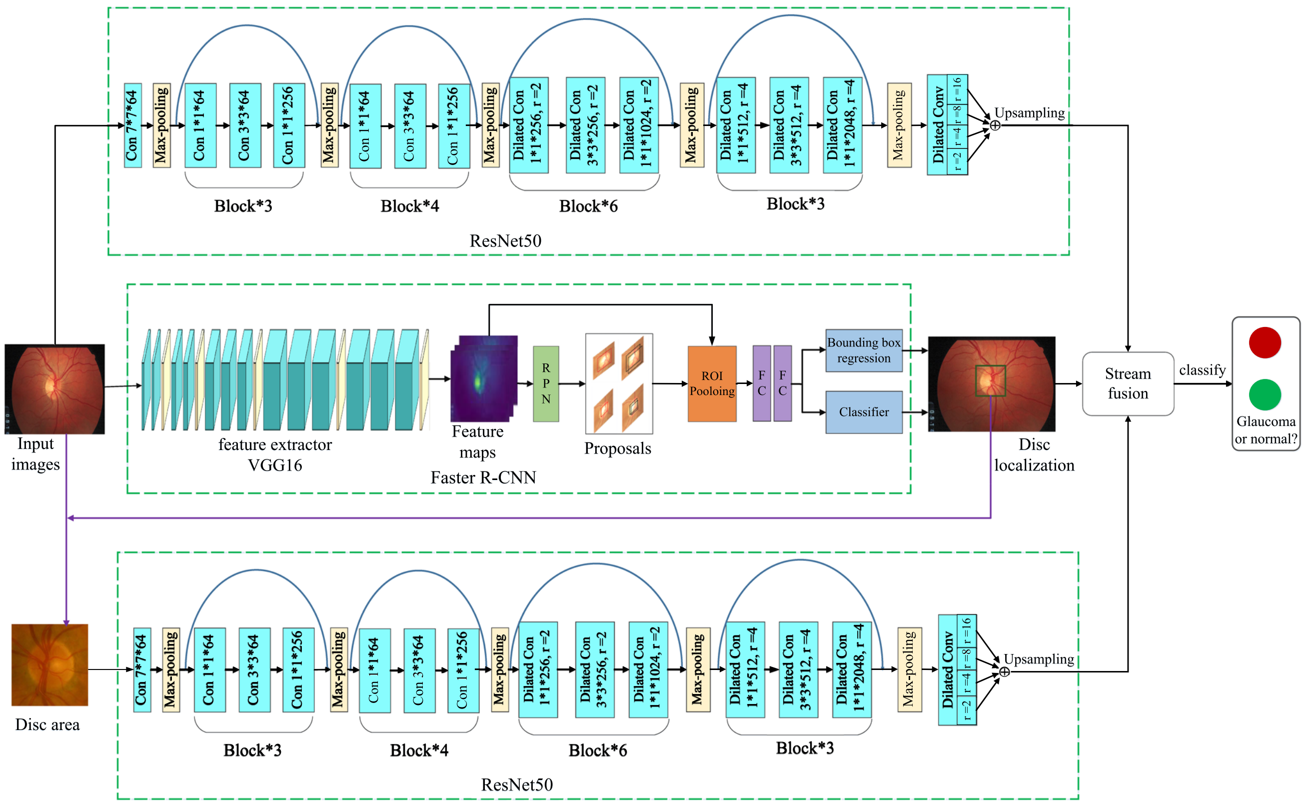

Fig. 4 Backbone framework of dual-tutor student learning model

connected layer to extract the rich multi-scale features of the optic disc, and the dilation rate is $\{2, 4, 8, 16\}$, respectively. Third, an up-sampling layer is connected to obtain the dense prediction results of the classification task.

Since the global fundus image has rich general structural information for guide accurate glaucoma detection, we resize the original fundus image input as 256×256 to 224×224 and input it into the modified ResNet50. Finally, two streams predictions of optic disk area and global fundus image are integrated to realize the classification of glaucoma and normal eyes.

Table 1 Overview of experimental datasets

Dataset	Total size	Healthy	Glaucoma	Country/race
LAG	4854	1711	3143	China/yellow
REFUGE	1200	1080	120	China/yellow
ORIGA	650	482	168	Singapore/yellow
Drishti-GS	101	31	70	India/mix
RIM-ONE-r1	169	118	40	Spain/white
RIM-ONE-r2	455	255	200	Spain/white
RIM-ONE-r3	159	85	74	Spain/white
ACRIMA	705	309	396	Spain/white
RIGA	750	–	–	Multi-source

4 Experiments and results

4.1 Datasets and scenarios

4.1.1 Datasets

We conducted experiments on nine public datasets from four countries: LAG [20], REFUGE [38], ORIGA [36], Drishti-GS [32], RIM-ONE (Release 1, 2, 3) [33], ACRIMA [39] and RIGA [40]. The general information of these dataset is shown in Table 1.

(1) LAG: This dataset contains 4854 fundus photography from Beijing Tongren Hospital, including 3143 glaucoma samples and 1711 healthy eyes with a resolution of 500×500. Each color fundus photography has a corresponding gold standard and attention map of ophthalmologist.

(2) REFUGE: This dataset contains 1200 groups of Chinese retinal fundus images. The ratio of glaucoma to normal eyes is 1:9. The dataset is split into 3 equal subsets for training, validation and testing. All three subsets provide both original images and clinical glaucoma labels. The resolution of training

images is 2124×2056, while the resolution of validation images and test images is 1634×1634.

(3) ORIGA: This dataset includes 650 fundus images collected from Singapore Malay Eye Study. The age of glaucoma patients ranges from 40 to 80 years. In the dataset, 482 are normal eyes and 168 are glaucoma.

(4) Drishti-GS: This dataset includes 101 Indian fundus retinal images with a resolution of 2047×1759. Each image includes glaucoma labels with qualified glaucoma experts.

(5) RIM-ONE-r1: This dataset is composed of 169 Spanish ONH images and corresponding image-level labels, which including 118 normal eyes, 40 glaucoma of varying degrees and 11 ocular hypertensions.

(6) RIM-ONE-r2: This dataset contains 455 color fundus images and glaucoma label by ophthalmologists, which including 255 healthy images and 200 glaucoma eyes.

(7) RIM-ONE-r3: This dataset contains 159 Spanish stereo fundus images with a resolution of 2144×1424 and manual annotations, which including 85 normal eyes and 74 glaucoma of varying degrees.

(8) ACRIMA: This dataset is composed by 705 fundus images collected from Spain, which including 309 normal images and 396 glaucomatous images. Each image includes manual labels by ophthalmologists with several clinical experiences. They were cropped around the optic disc and renamed.

(9) RIGA: This dataset includes 750 original images of various sizes and 4500 manual annotations of the optic disc/cup. But this dataset has no glaucoma label.

4.1.2 Scenarios

We consider the fundus image datasets of yellow, mixed, and white people as three different domains; thus, we built three different experimental scenarios to evaluate the domain adaptive performance of the proposed method.

(1) Yellow race $\rightarrow$ Caucasian. We used the training set of LAG, REFUGE, and ORIGA as the labeled source domain, RIGA as the unlabeled target domain, and the training set of RIM-ONE (Release 1,2,3) and ACRIMA as the labeled target domain. We evaluated the experimental performance on the RIM-ONE (Release 1,2,3) and ACRIMA test set.

(2) Caucasian $\rightarrow$ Mixed race. We used the training set of RIM-ONE (Release 1,2,3) and ACRIMA as S , RIGA as U, and the training set of DRISHTI-GS as T. The performance of the presented method was evaluated the on the test set of DRISHTI-GS.

(3) Mixed race $\rightarrow$ yellow race. We used the training set of DRISHTI-GS as S, RIGA as U, and the test sets of LAG, REFUGE and ORIGA as T. The experimental results were evaluated on the test sets of LAG, REFUGE and ORIGA datasets.

4.2 Implementation details

We experimented with the proposed method on an Ubuntu 18.04 system with an NVIDIA GeForce RTX 2080Ti graphics card with 11 GB of RAM. The network was implemented based on the PyTorch platform. The batch size is set to 4, that is, there are two images per domain. Faster R-CNN and ResNet50 used in the backbone network are initialized with parameters pre-trained on ImageNet [41]. We adopted the stochastic gradient descent (SGD) method to train the detection network and used the multi-learning strategy to update the learning rate. The learning rate was the initial learning rate multiplied by $(1 - \frac{iter}{max\_iter})^{power}$, where the power was 0.9, the initial learning rate was 10^{-4}, and the maximum number of iterations was 100. We used the Adam optimizer [42] to train the CycleGAN [9] with a learning rate of 10^{-5}. We set the balance parameters α_1 and α_2 in L^{cly}_{tutor} as 0.4 and 0.6, respectively, and set the balance factors η_1 and η_2 in $L^{cly}_{student}$ to 0.3 and 0.7, respectively. The exponential decay rate β of intra-image tutor model is set to 0.99. γ_1 is set to 0.1, and γ_2 changes according to the function $\gamma_2(c) = 0.1 \times e^{-5(1-c/c_{max})^2}$ in $L_{student}$, where c and c_{max} represent the current and final training steps, respectively, and c_{max} is set to 100 in the experiment.

Considering the problem of artifacts in training CycleGAN, we used the following two strategies to ensure the quality of synthetic images. First, RELU is adopted as the activation function of each layer in the generator, and LeakyReLU is used as the activation function of each layer in the discriminator. Second, in the decoding stage of the generator, bilinear interpolation up-sampling and convolution operations are used to replace the transposed convolution operations to reduce artifacts.

All the images in the experimental datasets were uniformly resized to 256×256 because the image sizes are inconsistent. In addition, the data augmentation needed to be performed due to the limited number of image samples. First, the input images were scaled randomly by a factor of [0.65, 1.25] in steps of 0.1. Second, the original images were rotated randomly within the range of $[0, 360°]$ in steps of 90°. Third, the original images were translated from -60 pixels to 120 pixels in 30-pixel steps horizontally and vertically. In the experiment, we randomly selected 70% glaucoma and normal eye images of datasets for training, and the remaining 30% images for testing. Different data augmentation operations were performed depending on the specific situations of each dataset. The distribution of training and test images is shown in Table 2.

4.3 Evaluation metrics

To evaluate the classification performances of different methods, we adopted the accuracy (*Acc*), sensitivity (*Sen*), specificity (*Spe*) as metrics, which are calculated as follows:

$$Acc = \frac{1}{2}(sen + spe), \tag{10}$$

$$Sen = \frac{TP}{TP + FN}, \tag{11}$$

$$Spe = \frac{TN}{TN + FP}, \tag{12}$$

where *TP* is true positive which indicates the number of glaucoma images that correctly predicted by the network. *FP* is false positive which represents the number of normal eyes misclassified as glaucoma. *TN* is true negative which indicates the number of normal images that correctly classified. *FN* is false negative which denotes the number of glaucomatous images misclassified as healthy images. Furthermore, we used the receiver operating characteristic (ROC) curve to visualize the performances of different methods for glaucoma screening.

Moreover, we used normalized cross correlation (NCC) and peak signal-to-noise ratio (PSNR) to evaluate the quality of synthetic target images from CycleGAN. The metrics of NCC and PSNR are calculated as follows:

$$NCC = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_i (X_i - \bar{X})^2 \sum_i (Y_i - \bar{Y})^2}}, \tag{13}$$

$$PSNR = 10lg\left(\frac{(2^n - 1)^2}{MSE}\right), \tag{14}$$

$$MSE = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} (X(i,j) - Y(i,j))^2, \tag{15}$$

where X_i and Y_i represent the RGB values of the two reference images, respectively. $\bar{X}$ and $\bar{Y}$ represent the average pixel values of real images and synthetic images, respectively. *MSE* represents the mean square error of the two reference images *X* and *Y*, while *H* and *W* represent the height and width of the fundus images. *n* represents the number of pixels in bits, which is set to 8, i.e., the number of pixel gray levels is 256.

4.4 Comparisons with state-of-the-art methods

4.4.1 Performance on fundus images of yellow race to Caucasian

To demonstrate the effectiveness of the proposed method, we compared the experimental performance of our method with nine state-of-the-art methods [5, 6, 13, 20, 21, 23, 25, 43, 44] on RIM-ONE (Release 1, 2, 3) and ACRIMA. Furthermore, we also compared the final results of our method with those using some components. The quantitative results are shown in Tables 3 and 4.

It can be seen from Tables 3 and 4 that the scores of our method are higher than those of most other methods, and the accuracy on RIM-ONE-r1, RIM-ONE-r2, RIM-ONE-r3 and ACRIMA are 0.9748. 0.9746, 0.9646 and 0.9652, respectively. Compared with Wu et al. [5], the accuracy is increased by 2.45%, 2.22%, 2.17% and 2.04%, respectively. In addition, the experimental results using DSLN model are better than those of the no-tutor and single-tutor model. The main reason is that inter-image tutor uses CycleGAN [9] to align the appearance from the labeled source domain to the labeled target domain and learn the

Table 2 Distribution of images used in the experiment

Dataset	Data augmentation	Number of training samples	Number of test samples
LAG	Not applied	G-2200 N-1197 T-3397	G-943 N-514 T-1457
REFUGE	Applied	G-320 N-2880 T-3200	G-160 N-1440 T-1600
ORIGA	Applied	G-2024 N-705 T-2729	G-868 N-303 T-1171
DRISHTI-GS	Applied	G-520 N-1176 T-1696	G-224 N-504 T-728
RIM-ONE-r1	Applied	G-1982 N-672 T-2654	G-850 N-288 T-1138
RIM-ONE-r2	Applied	G-1071 N-840 T-1911	G-459 N-360 T-819
RIM-ONE-r3	Applied	G-1428 N-1243 T-2671	G-612 N-533 T-1145
ACMIRA	Applied	G-1297 N-1663 T-2960	G-557 N-713 T-1270
RIGA	Applied	T-8400	T-3600

Note: G indicates glaucoma and N denotes normal images, while T represents total images

Table 3 Quantitative result comparison of different methods on the RIM-ONE (Release 1, 2) dataset

Methods	RIM-ONE-r1				RIM-ONE-r2			
	Acc	Sen	Spe	AUC	Acc	Sen	Spe	AUC
Wu et al. [5]	0.9503 ± 0.0213	0.9668 ± 0.0015	0.9357 ± 0.0119	0.9617	0.9524 ± 0.0211	0.9571 ± 0.0215	0.9256 ± 0.0214	0.9590
Pinto et al. [6]	0.8565 ± 0.0115	0.7986 ± 0.0221	0.8294 ± 0.0128	0.9165	0.8597 ± 0.1103	0.8175 ± 0.1023	0.8931 ± 0.1016	0.9027
Claro et al. [13]	0.9431 ± 0.0113	0.9047 ± 0.0105	0.9516 ± 0.0121	0.9582	0.9475 ± 0.0051	0.9282 ± 0.0120	0.9543 ± 0.0116	0.9623
Li et al. [20]	0.9332 ± 0.1028	0.9415 ± 0.0126	0.9512 ± 0.0217	0.9602	0.9521 ± 0.1019	0.9468 ± 0.1024	0.9535 ± 0.1123	0.8091
Fu et al. [21]	0.8429 ± 0.0124	0.8478 ± 0.0116	0.8380 ± 0.1048	0.8317	0.8497 ± 0.0121	0.8173 ± 0.1019	0.7876 ± 0.1137	0.8465
Chai et al. [23]	0.9151 ± 0.1025	0.9232 ± 0.0227	0.9190 ± 0.0218	0.9137	0.9102 ± 0.1032	0.9158 ± 0.0237	0.9134 ± 0.0232	0.9092
Bajwa et al. [25]	0.7897 ± 0.1125	0.7117 ± 0.1104	0.7751 ± 0.1037	0.8681	0.7889 ± 0.1012	0.7255 ± 0.1239	0.7543 ± 0.1121	0.8742
Elangovan et al. [43]	0.8591 ± 0.1052	0.8143 ± 0.1325	0.8867 ± 0.1028	0.8958	0.8597 ± 0.0136	0.8175 ± 0.0124	0.8931 ± 0.0132	0.9072
Raghavendra et al. [44]	0.9713 ± 0.0014	0.9687 ± 0.0025	0.9782 ± 0.0052	0.9277	0.9681 ± 0.0005	0.9622 ± 0.0126	0.9573 ± 0.0107	0.9549
No tutor	0.9562 ± 0.0025	0.9756 ± 0.0121	0.9634 ± 0.0120	0.9615	0.9676 ± 0.0026	0.9728 ± 0.0132	0.9615 ± 0.0021	0.9624
Inter-image tutor	0.9647 ± 0.0058	0.9728 ± 0.0127	0.9671 ± 0.0053	0.9741	0.9725 ± 0.0108	0.9756 ± 0.0124	0.9643 ± 0.0115	0.9619
Intra-image tutor	0.9652 ± 0.0107	0.9649 ± 0.0112	0.9633 ± 0.0113	0.9732	0.9681 ± 0.0053	0.9742 ± 0.0124	0.9589 ± 0.0026	0.9624
Proposed	**0.9748 ± 0.0101**	**0.9831 ± 0.0052**	**0.9714 ± 0.0027**	**0.9760**	**0.9746 ± 0.0126**	**0.9823 ± 0.0023**	**0.9657 ± 0.0024**	**0.9640**

The best results are shown in bold

prior knowledge of the source domain, while intra-image tutor uses the EMA of the student network to utilize the potential knowledge from the unlabeled target domain. Two tutor networks transfer learned knowledge to the student model, which fully integrate the advantages of domain adaptation and semi-supervised learning.

However, Fu et al. [21] do not fully consider the fundus structure differences between yellow and white people, and only evaluate the performance on two private fundus image datasets of Chinese and Indians. Therefore, the experimental performance on the fundus images of Caucasians still has room for improvement. The experimental results on RIM-ONE (Release 1, 2) show that the method proposed by Li et al. [20] is susceptible to the influence of the physiological large cupping to produce false positive results. Furthermore, the method presented by Bajwa et al. [25] is difficult to learn discriminative features for glaucoma detection in public datasets. The main reason is that as the depth of network deepens, the fine-grained distinguishing features of fundus images gradually disappear, making it difficult to accurately detect glaucoma images in public datasets.

Figure 5 shows the ROC curves of the five methods on RIM-ONE-r1, RIM-ONE-r2, RIM-ONE-r3 and ACRIMA. The closer the ROC curve is to the upper left boundary, the more accurate the trained model is. It is observed from Fig. 5a that the ROC curve of our proposed method is the upper-left curve of the five models, and the curve of DENet [21] is the lowest curve of the five curves. The data in the lower right corner of Fig. 5d show that the AUC value of our method is the largest, followed by Chai et al. [23], and Fu et al. [21] is the smallest.

Figure 6 displays the correct classification results of the proposed DSLN on the RIM-ONE-r1, RIM-ONE-r2, RIM-ONE-r3 and ACRIMA datasets from top to bottom. Cup-to-disk ratio (CDR) is often used as an important basis for glaucoma detection. As shown in Fig. 6, the purple and blue ellipses represent optic cup (OC) and optic disc (OD), respectively. CDR refers to the ratio between the vertical cup diameter (VCD) and the vertical disc diameter (VDD). A larger CDR means a higher risk of glaucoma. It is

Table 4 Quantitative result comparison of different methods on the RIM-ONE-r3 and ACRIMA datasets

Methods	RIM-ONE-r3				ACMIRA			
	Acc	Sen	Spe	AUC	Acc	Sen	Spe	AUC
Wu et al. [5]	0.9429 ± 0.0105	0.9623 ± 0.0121	0.9347 ± 0.0108	0.9600	0.9448 ± 0.0025	0.9576 ± 0.0141	0.9265 ± 0.0109	0.9110
Pinto et al. [6]	0.8651 ± 0.0123	0.8017 ± 0.0126	0.8296 ± 0.0107	0.8988	0.9664 ± 0.0105	0.9607 ± 0.0109	0.9739 ± 0.0121	0.9014
Claro et al. [13]	0.9457 ± 0.0103	0.9432 ± 0.0107	0.9457 ± 0.0059	0.9728	0.9531 ± 0.0132	0.9486 ± 0.0117	0.9523 ± 0.0102	0.9676
Li et al. [20]	0.8522 ± 0.0119	0.8468 ± 0.0123	0.8545 ± 0.0058	0.9366	0.9328 ± 0.0121	0.9417 ± 0.0057	0.9465 ± 0.0142	0.9512
Fu et al. [21]	0.8462 ± 0.0108	0.8371 ± 0.0109	0.8216 ± 0.0121	0.8147	0.8452 ± 0.0205	0.8068 ± 0.0124	0.7796 ± 0.0217	0.8368
Chai et al. [23]	0.9126 ± 0.0103	0.9214 ± 0.0108	0.9087 ± 0.0125	0.9147	0.9232 ± 0.0107	0.9181 ± 0.0003	0.9045 ± 0.0047	0.9170
Bajwa et al. [25]	0.7862 ± 0.0105	0.7274 ± 0.0118	0.7652 ± 0.0241	0.8713	0.7852 ± 0.0103	0.7129 ± 0.0126	0.7632 ± 0.0217	0.8645
Elangovan et al. [43]	0.8562 ± 0.0107	0.8235 ± 0.0113	0.8754 ± 0.0120	0.8984	0.8597 ± 0.0142	0.8175 ± 0.0108	0.8931 ± 0.1102	0.9002
Raghavendra et al. [44]	**0.9673 ± 0.0051**	0.9625 ± 0.0147	**0.9733 ± 0.0092**	0.9173	0.9624 ± 0.0113	0.9681 ± 0.0109	**0.9742 ± 0.0118**	0.9356
No tutor	0.9516 ± 0.0103	0.9594 ± 0.0056	0.9513 ± 0.0178	0.9572	0.9576 ± 0.0102	0.9581 ± 0.0209	0.9552 ± 0.0121	0.9645
Inter-image tutor	0.9624 ± 0.0102	0.9671 ± 0.0031	0.9652 ± 0.0106	0.9613	0.9622 ± 0.0132	0.9664 ± 0.0069	0.9647 ± 0.0041	0.9686
Intra-image tutor	0.9572 ± 0.0117	0.9651 ± 0.0102	0.9583 ± 0.0107	0.9636	0.9634 ± 0.0114	0.9583 ± 0.0106	0.9576 ± 0.0057	0.9691
Proposed	0.9646 ± 0.0102	**0.9718 ± 0.0051**	0.9682 ± 0.0062	**0.9656**	**0.9652 ± 0.0113**	**0.9762 ± 0.0021**	0.9672 ± 0.0014	**0.9726**

The best results are shown in bold

observed from Fig. 6 that our method can overcome the influence of blood vessel interference around the optic disc and low contrast, and can accurately detect glaucoma when the appearance of fundus images is quite different from the LAG, ORIGA, and REFUGE datasets. The prediction results show that our method has high robustness and accuracy.

Figure 7a ~ d displays the visualized results of the inter-image tutor model with LAG as the source domain, ACRIMA as the target domain, using CycleGAN [9] for appearance alignment. Figure 7e ~ h shows the visualized samples of the inter-image tutor model with ORIGA as the source domain, RIM-ONE-r2 as the target domain, using CycleGAN [9] for style transfer. It can be seen from Fig. 7 that the synthetic fundus images not only preserve the detailed features of original image, but also successfully transformed into another style of appearance. Moreover, Table 5 shows the quantitative results of the synthetic target image quality in different domain shift scenarios. The NCC and PSNR in Table 5 are the average values of all images in the target domain dataset. It can be seen from Table 5 that the fake target images generated by CycleGAN are of good quality and have higher similarity compared with the source domain images.

4.4.2 Performance on fundus images of Caucasian to mixed race

Table 6 shows the quantitative comparison results between the proposed method and the other existing methods on the DRISHTI-GS test set. It is observed from Table 6 that the experimental results of our method are superior to those of the current state-of-the-art methods, and the proposed DSLN framework reaches values of 0.9845, 0.9766, 0.9684 and 0.9704 on Acc, Sen, Spe and AUC, respectively. The main reason is that we take the teacher-student model as the basic framework and use two teacher models to train the network. The inter-image tutor uses CycleGAN [9] to realize the style transfer from the source domain image to the target domain image to narrow the appearance difference between different domains, and transfer the prior knowledge learned from the labeled source domain to the

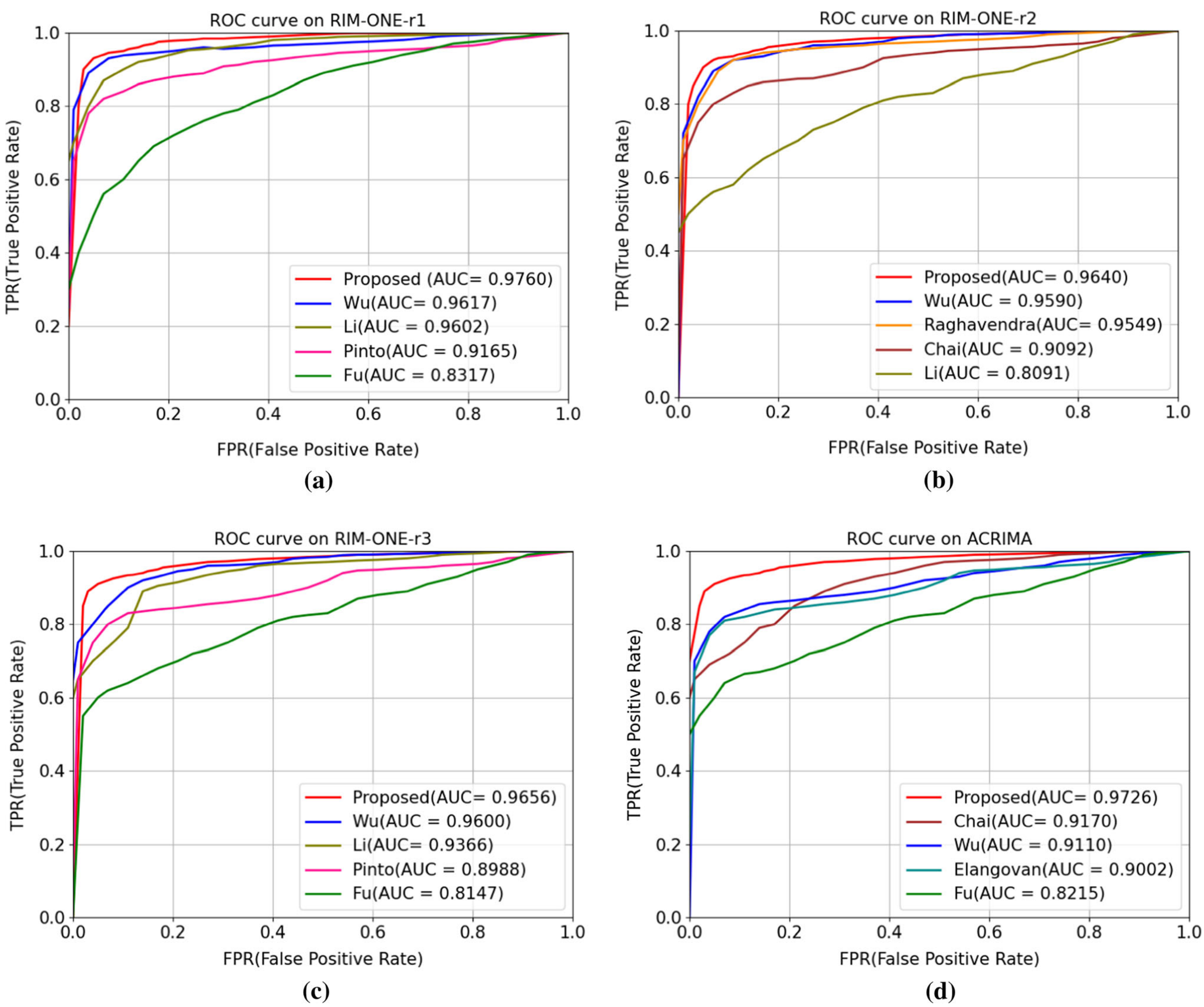

Fig. 5 ROC curves on four datasets for glaucoma screening **a** ROC curve on RIM-ONE-r1; **b** ROC curve on RIM-ONE-r2; **c** ROC curve on RIM-ONE-r3; **d** ROC curve on ACRIMA

student model. The intra-image tutor updates the training process according to EMA of the student model and transfers the learned knowledge from the unlabeled target domain to the student model. The student network not only acquires knowledge directly from the labeled target domain images, but also obtains the inter-image knowledge and intra-image knowledge from two tutors. The purpose of the proposed method is to effectively combine the advantages of domain adaptation and semi-supervised learning. In addition, we use the modified ResNet50 to extract the multi-scale features from the optic disc area and the whole image and merge them as the backbone network for glaucoma classification. Although the appearance of fundus images of mixed races and Caucasians is quite different, the proposed DSLN can still detect glaucoma images accurately. Experimental results show that our method can effectively overcome the domain shift and has good robustness.

However, Wu et al. [5] only use one teacher model to learn knowledge from the undiagnosed images, and ignores the appearance and feature differences between the undiagnosed images and the diagnosed images, which results in poor robustness. Pinto et al. [6] paid more attention to the information of the optic disc area, ignoring the global image feature representation, which is easy to misclassify the healthy image as a glaucoma. Moreover, Chai et al. [23] can make full use of image features and non-image features, but it relies on many labeled images and clinical data of patients to train the network for ensuring the detection accuracy, which is difficult to generalize to real clinical applications.

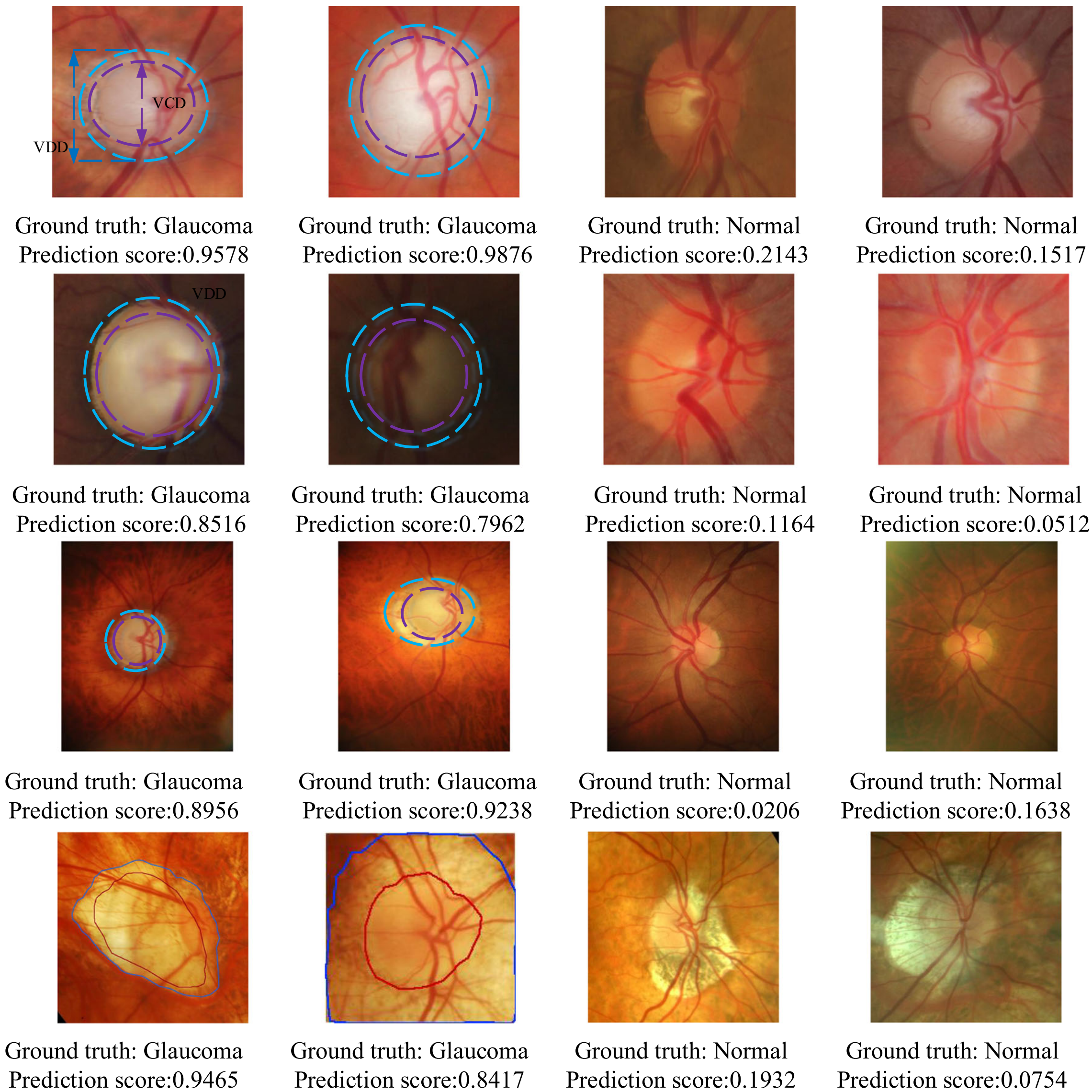

Fig. 6 Correct classification results of glaucoma and normal eyes on RIM-ONE (Release 1, 2, 3) and ACRIMA. The purple and blue lines represent detected lesion areas of glaucoma

Figure 8 shows the ROC curves of the five methods on DRISHTI-GS. It can be seen from Fig. 8 that the ROC curve of our proposed method is the upper-left curve while the curve of DCGAN [6] is the lowest curve of the five curves. The data in the lower right corner of Fig. 8 shows that the AUC value of our method is the largest, followed by Raghavendra et al. [44], and Pinto et al. [6] is the smallest. Compared with Pinto et al. [6], the AUC of our method is increased by 14.98%.

Figure 9 displays the correct classification results of our method on the DRISHTI-GS dataset. It can be seen from Fig. 9 that there are macula and intertwined blood vessels around the optic disc area of the fundus image. We use Faster R-CNN to accurately locate the optic disc area and extract multi-scale context features from both the global image and the local optic disc area to achieve glaucoma detection. The experimental results show that the proposed method can effectively overcome the lesion noise and vascular interference, and accurately classify glaucoma and normal eyes.

To verify the robustness of our method, we added Gaussian noise with different variances to the original image of DRISHTI-GS for glaucoma detection. Figure 10 shows fundus image samples of Gaussian noise with a variance of 0.5. It is observed that the images with Gaussian noise becomes blurred, and it is difficult to distinguish the structural features. Table 7 shows the quantitative results in the presence of Gaussian noise with different variances. It can be seen from Table 7 that as the variance of Gaussian noise increases from 0.1 to 0.5, the accuracy of

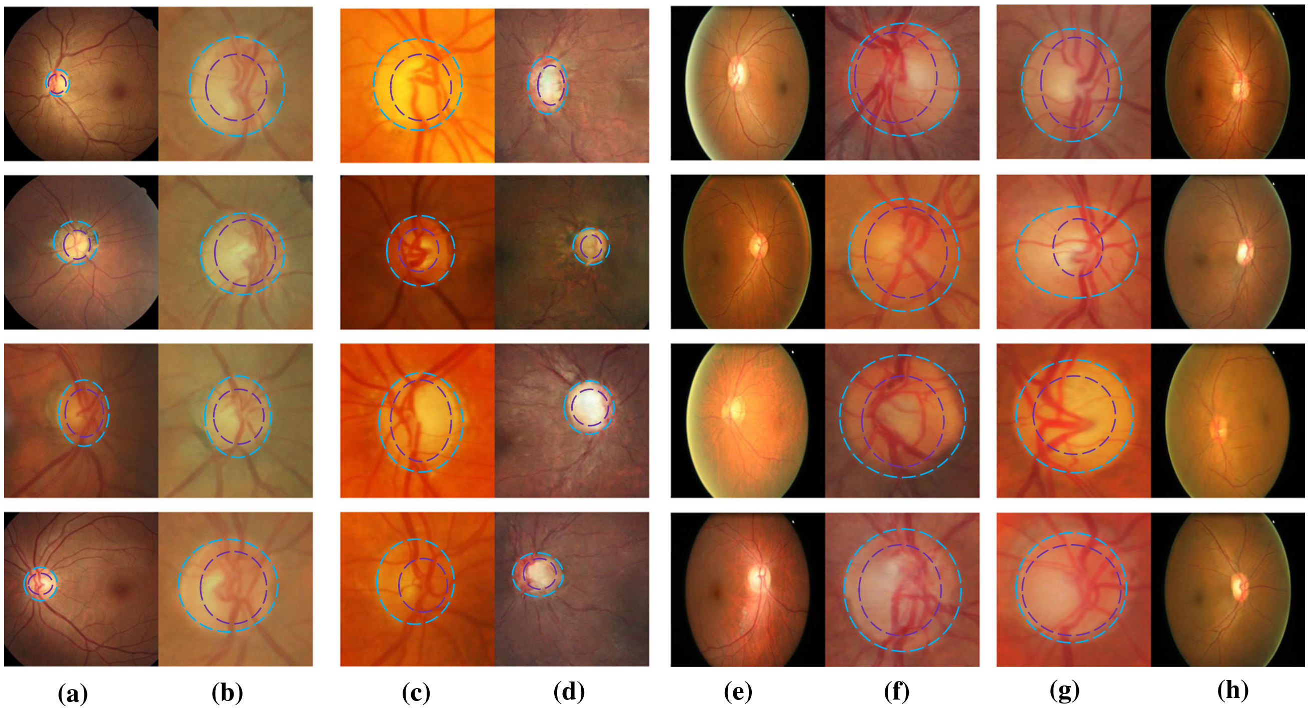

Fig. 7 Image style transfer results between LAG and ACRIMA datasets, ORIGA and RIM-ONE-r2 datasets **a** original fundus images in LAG **b** style transfer results from LAG to ACRIMA **c** original fundus images in ACRIMA **d** style transfer results from ACRIMA to LAG **e** original fundus images in ORIGA **f** style transfer results from ORIGA to RIM-ONE-r2 **g** original fundus images in RIM-ONE-r2 **h** style transfer results from RIM-ONE-r2 to ORIGA

Table 5 Quality assessment by NCC and PSNR on fundus images of yellow race to Caucasian

Scenarios	NCC	PSNR
LAG→ACRIMA	0.58	34.57
ORIGA→RIM-ONE-r2	0.62	35.62
REFUGE→RIM-ONE-r3	0.51	29.83

glaucoma detection decreases from 0.9529 to 0.9457. In addition, we compare the proposed method with other methods on fundus images with Gaussian noise with variances of 0.2 and 0.5. The accuracy comparison results are shown in Fig. 11. Compared with other methods, the proposed DSLN is less susceptible to interference from Gaussian noise, and still maintains a relatively stable accuracy in the presence of obvious Gaussian noise. The experimental results show that our method can adapt to the changes of Gaussian noise and has strong robustness.

4.4.3 Performance on fundus images of mixed race to yellow race

We compared the experimental performance of our proposed DSLN model with other state-of-the-art methods on the test sets of LAG, ORIGA and REFUGE, and the quantitative results are shown in Tables 8 and 9. It is observed from Tables 8, 9 that the performances of the proposed DSLN are better than those of other methods, and the accuracy on LAG, ORIGA and REFUGE are 0.9814, 0.9764 and 0.9874, respectively. Compared with the other two datasets, the performances of the proposed DSLN on LAG is better. Moreover, the experimental results of the DSLN are better than those of the no-tutor, the intra-image tutor and inter-image tutor model. The main reason is that the student model not only acquires intra-image knowledge and inter-image knowledge from the two tutor models, but also obtains knowledge from the labeled target domain. Moreover, our backbone network uses a modified ResNet50 to capture feature representations from both the global fundus images and the local optic disk areas, and fully integrates the features extracted from the two streams for accurate glaucoma detection. Although there are obvious differences in the appearance and feature distribution of fundus images between mixed races and yellow races, our method has good robustness and generalization: it can

Table 6 Quantitative result comparison of different methods on the DRISHTI-GS dataset

Methods	DRISH-GS			
	Acc	Sen	Spe	AUC
Wu et al. [5]	0.9495 ± 0.0052	0.9652 ± 0.0107	0.9357 ± 0.0114	0.9097
Pinto et al. [6]	0.8451 ± 0.0113	0.7986 ± 0.0105	0.8291 ± 0.0109	0.8206
Claro et al. [13]	0.9552 ± 0.0112	0.9154 ± 0.0109	0.9547 ± 0.0113	0.9715
Li et al. [20]	0.9452 ± 0.0102	0.8458 ± 0.0107	0.8876 ± 0.0051	0.8938
Fu et al. [21]	0.8428 ± 0.0107	0.8573 ± 0.0059	0.8382 ± 0.0032	0.8521
Chai et al. [23]	0.9153 ± 0.0056	0.9237 ± 0.0121	0.9098 ± 0.0107	0.9147
Bajwa et al. [25]	0.7794 ± 0.0214	0.7938 ± 0.0116	0.7785 ± 0.0113	0.8714
Elangovan et al. [43]	0.8662 ± 0.0103	0.9231 ± 0.0124	0.4815 ± 0.0117	0.9012
Raghavendra et al. [44]	0.9713 ± 0.0108	0.9605 ± 0.0117	0.9623 ± 0.0126	0.9119
No tutor	0.9625 ± 0.0102	0.9647 ± 0.0105	0.9538 ± 0.0061	0.9547
Inter-image tutor	0.9789 ± 0.0121	0.9657 ± 0.0107	0.9645 ± 0.0104	0.9638
Intra-image tutor	0.9752 ± 0.0052	0.9761 ± 0.0047	0.9625 ± 0.0124	0.9643
Proposed	**0.9845 ± 0.0106**	**0.9766 ± 0.0029**	**0.9684 ± 0.0043**	**0.9704**

The best results are shown in bold

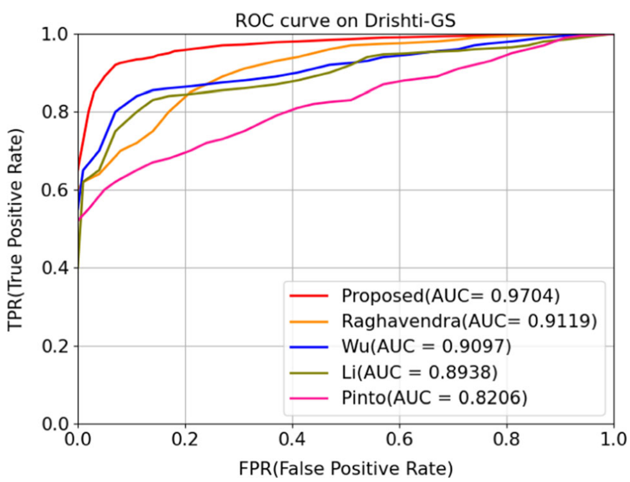

Fig. 8 ROC curves of different methods for glaucoma detection on DRISHTI-GS

overcome the influence of domain shift and detect glaucoma accurately under the condition of low contrast and vascular interference. But the model of Elangovan et al. [43] does not work well on database with unbalanced categories. Raghavendra et al. [44] does not consider the difference in ONH appearance of mixed races and yellow races, which makes it difficult for models trained on DRISHTI-GS to get better results on LAG and ORIGA.

Figure 12 shows the ROC curves of the five methods on LAG, ORIGA and REFUGE. It can be seen from Fig. 12b that the ROC curve of the proposed DSLN is the upper-left curve of the five models, while the curve of Fu et al. [21] is the lowest curve among the five curves. The data in the lower right corner of Fig. 12c shows that the AUC value of our method is the largest, followed by Li et al. [20], and Bajwa [25] is the smallest. Compared with Bajwa [25], the AUC of our method is increased by 13.81%.

Figure 13 shows the correct classification results of our method on the LAG, ORIGA and REFUGE datasets from top to bottom. It is observed that there are the obvious dark spots in the fundus image of LAG, and the fundus image in ORIGA has low contrast and macular noise, which may affect the accuracy of glaucoma classification. They are obviously different from the images of DRISHTI-GS in appearance and feature distribution. However, the proposed DSLN adopts two tutors to fully learn the rich features of the labeled source images and unlabeled target images, which can effectively overcome the adverse effects of domain shift and realize accurate glaucoma detection.

Figure 14 shows the training and validation loss curve on the ACRIMA, DRISHTI-GS, LAG and REFUGE datasets to illustrate the convergence performance of the proposed model. It can be observed from Fig. 14 that with the increase of the number of network epochs, the training loss and validation loss gradually decrease and tend to stabilize from the 20th epoch. It indicates that the proposed method has better convergence performance.

In summary, the experimental results on fundus images of Caucasians, yellow races and mixed races demonstrate that the proposed DSLN framework has better results on REFUGE and LAG. There are two main reasons. First, the fundus images of LAG and REFUGE contain both global fundus structural features and detailed features of the local optic disc area, making the features extracted by the backbone network more comprehensive and richer, which is conducive to accurate detection of glaucoma. However, the original images of RIM-ONE (Release 1, 2, 3) and ACRIMA are cropped optic disc regions that do not

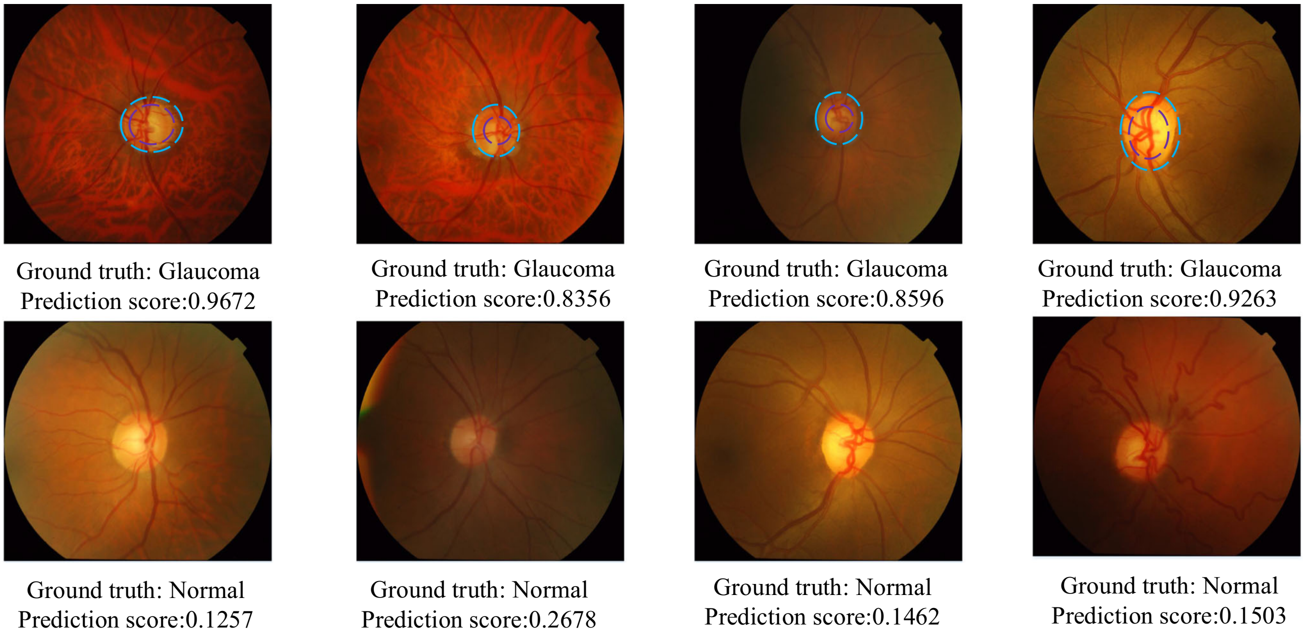

Fig. 9 Correct classification results of glaucoma and normal eyes on DRISHTI-GS. The purple and blue ellipses represent detected lesion areas of glaucoma

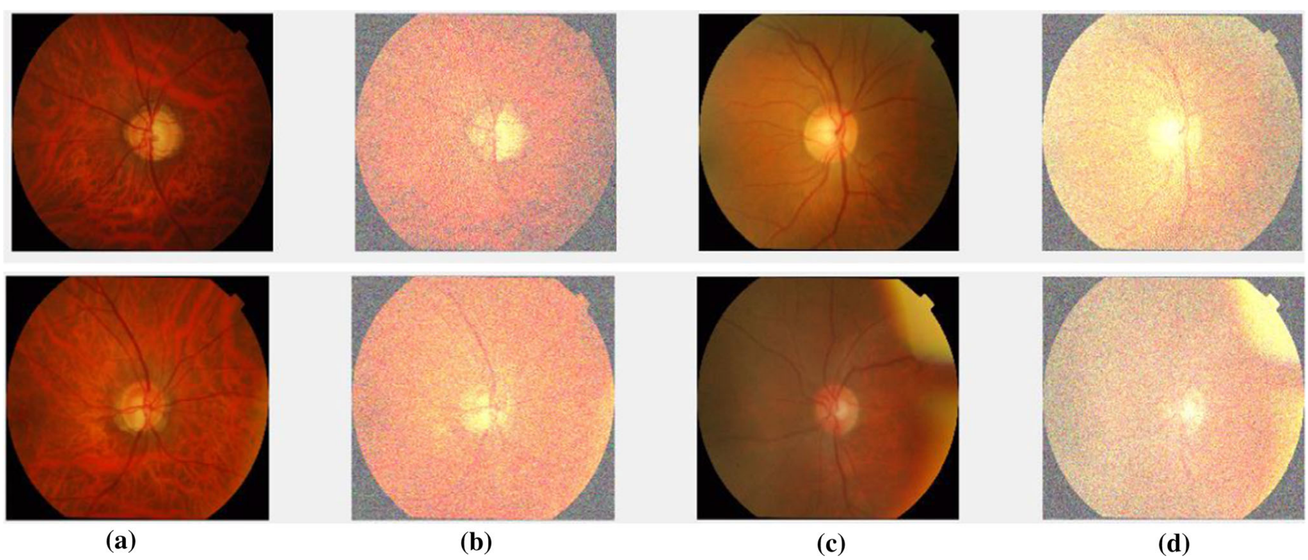

Fig. 10 Examples of glaucoma and normal images with Gaussian noise in the DRISHTI-GS dataset **a** original glaucoma images; **b** glaucoma images of Gaussian noise with variance 0.5; **c** original normal images; **d** normal images of Gaussian noise with variance 0.5

Table 7 Quantitative results on DRISHTI-GS with Gaussian noise

Variance	Acc	Sen	Spe
0.1	0.9529 ± 0.0051	0.9716 ± 0.0026	0.9658 ± 0.0121
0.2	0.9518 ± 0.0025	0.9698 ± 0.0125	0.9643 ± 0.0027
0.3	0.9487 ± 0.0023	0.9673 ± 0.0047	0.9632 ± 0.0106
0.4	0.9462 ± 0.0027	0.9656 ± 0.0052	0.9628 ± 0.0109
0.5	0.9457 ± 0.0046	0.9641 ± 0.0107	0.9617 ± 0.0105

include global fundus features. Therefore, the detection performance is not as good as those on LAG and REFUGE. Second, the number of image samples in the LAG and REFUGE datasets is obviously more than that in other datasets. The abundant data are beneficial to improve the accuracy and robustness of training network. Therefore, the experimental performances of our method on these two datasets are better than those on other datasets.

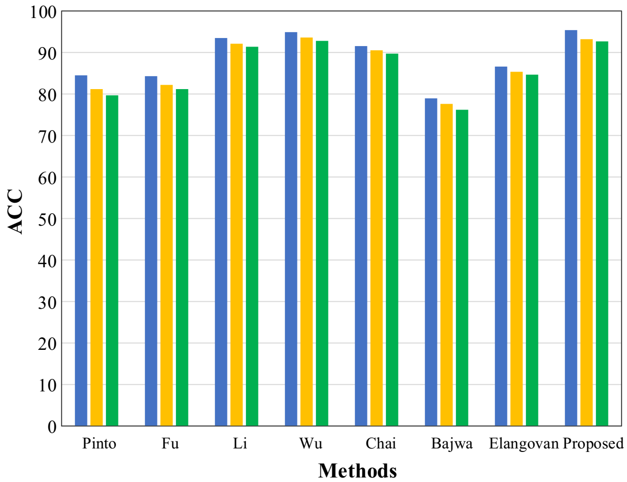

Fig. 11 Accuracy comparison of different methods in the presence of Gaussian noise on DRISHTI-GS

4.5 Ablation study

To verify the effectiveness of each component in our approach and its contribution to the results, we performed the following ablation study on LAG, ACRIMA and DRISHTI-GS, and adopted *Acc* and *AUC* to evaluate the performance of each method.

The ablation studies mainly consist of three parts: no-tutor network, one-tutor network and two-tutor network. First, we jointly train the backbone network on the labeled source images and the labeled target images, and use the pseudo-label approach [45] for the unlabeled target images, and then retrain the backbone on the three domain images, which is used as the baseline. In addition, we use Cycle-GAN [9] for transfer style from the source domain S to the synthetic target domain $D^{s\to t}$ to reduce the appearance difference between different domain and use the pseudo-label method [45] for unlabeled target domain U, and then jointly train the backbone on $D^{s\to t}$ and U, which is used as the GAN baseline. Neither the baseline nor the GAN baseline uses the teacher-student network scheme. Second, we experiment with the one tutor model. In the absence of intra-image tutor, we use the pseudo-label method [45] in the unlabeled target domain for network training. In the absence of inter-image tutor, we use CycleGAN [9] to align the appearance of the source images and target images, and

Table 8 Quantitative comparison results on the LAG and ORIGA datasets

Methods	LAG				ORIGA			
	Acc	Sen	Spe	AUC	Acc	Sen	Spe	AUC
Wu et al. [5]	0.9604 ± 0.0103	0.9872 ± 0.0102	0.9475 ± 0.0016	0.9127	0.9364 ± 0.0056	0.9637 ± 0.0031	0.9182 ± 0.0019	0.9081
Pinto et al. [6]	0.8152 ± 0.0114	0.7896 ± 0.0126	0.8394 ± 0.0058	0.8886	0.8234 ± 0.0103	0.7986 ± 0.0105	0.8368 ± 0.0130	0.9153
Li et al. [20]	0.9534 ± 0.0020	0.9544 ± 0.0015	0.9523 ± 0.0103	0.9754	0.9463 ± 0.1041	0.9521 ± 0.0025	0.9467 ± 0.0018	0.9184
Fu et al. [21]	0.8462 ± 0.0012	0.8371 ± 0.0203	0.8265 ± 0.0015	0.8284	0.8152 ± 0.0106	0.8068 ± 0.0047	0.7896 ± 0.0024	0.8306
Chai et al. [23]	0.9142 ± 0.0023	0.9236 ± 0.0105	0.9067 ± 0.0103	0.9158	0.9236 ± 0.0102	0.9142 ± 0.0107	0.8957 ± 0.0028	0.8851
Bajwa et al. [25]	0.8231 ± 0.0014	0.9186 ± 0.0056	0.8687 ± 0.1080	0.8859	0.7897 ± 0.1217	0.7983 ± 0.2061	0.8521 ± 0.0119	0.8936
Elangovan et al. [43]	0.9443 ± 0.0017	0.9476 ± 0.0026	0.9426 ± 0.0025	0.9134	0.7832 ± 0.0108	0.5806 ± 0.0113	0.9244 ± 0.0015	0.9251
Raghavendra et al. [44]	0.9574 ± 0.0025	0.9415 ± 0.0047	0.9756 ± 0.0021	0.9195	0.9543 ± 0.0028	0.9675 ± 0.0019	0.9728 ± 0.0008	0.9587
No tutor	0.9658 ± 0.0107	0.9761 ± 0.0052	0.9728 ± 0.0028	0.9627	0.9654 ± 0.0028	0.9628 ± 0.0131	0.9645 ± 0.0102	0.9655
Inter-image tutor	0.9717 ± 0.0102	0.9843 ± 0.0025	0.9794 ± 0.0108	0.9638	0.9725 ± 0.0105	0.9691 ± 0.0058	0.9713 ± 0.0121	0.9681
Intra-image tutor	0.9763 ± 0.0054	0.9824 ± 0.0028	0.9735 ± 0.0047	0.9619	0.9743 ± 0.0105	0.9674 ± 0.0107	0.9682 ± 0.0113	0.9616
Proposed	**0.9814 ± 0.0015**	**0.9862 ± 0.0023**	**0.9817 ± 0.0014**	**0.9641**	**0.9764 ± 0.0026**	**0.9713 ± 0.0016**	**0.9762 ± 0.0050**	**0.9693**

The best results are shown in bold

Table 9 Quantitative comparison results on the REFUGE dataset

Methods	REFUGE			
	Acc	Sen	Spe	AUC
Wu et al. [5]	0.8061 ± 0.0102	0.8291 ± 0.0045	0.9034 ± 0.0029	0.9058
Pinto et al. [6]	0.8392 ± 0.0105	0.8497 ± 0.0021	0.8386 ± 0.0053	0.8465
Claro et al. [13]	0.9425 ± 0.0019	0.9472 ± 0.0023	0.9516 ± 0.0108	0.9175
Li et al. [20]	0.9587 ± 0.0026	0.9623 ± 0.0029	0.9451 ± 0.0005	0.9128
Fu et al. [21]	0.9134 ± 0.0104	0.9216 ± 0.0106	0.9047 ± 0.0108	0.8881
Chai et al. [23]	0.8851 ± 0.0056	0.9134 ± 0.0121	0.9098 ± 0.0107	0.9147
Bajwa et al. [25]	0.7953 ± 0.0152	0.7987 ± 0.0129	0.8296 ± 0.0116	0.8325
Elangovan et al. [43]	0.9452 ± 0.0103	0.9454 ± 0.0071	0.9417 ± 0.0032	0.9532
Raghavendra et al. [44]	0.9697 ± 0.0122	0.9653 ± 0.0028	**0.9668 ± 0.0023**	0.9684
No tutor	0.9624 ± 0.0018	0.9573 ± 0.0056	0.9548 ± 0.0107	0.9653
Inter-image tutor	0.9645 ± 0.0012	0.9582 ± 0.0106	0.9632 ± 0.0032	0.9638
Intra-image tutor	0.9735 ± 0.0114	0.9584 ± 0.0058	0.9517 ± 0.0027	0.9702
Proposed	**0.9874 ± 0.0071**	**0.9685 ± 0.0052**	0.9657 ± 0.0103	**0.9706**

The best results are shown in bold

jointly train the backbone network on the synthesized target images and the labeled target images. Third, we experiment with the proposed DSLN framework. The ablation results are shown in Table 10.

It can be seen from Table 10 that the baseline does not acquire knowledge transfer from the tutor model, and the knowledge in the labeled source image and the unlabeled target image is not fully utilized, so the experimental performance is poor. The GAN baseline uses CycleGAN [9] to reduce the appearance difference between the source images and the target images and thus its performance is better than the baseline, but there is still a gap compared with the performance of the network using the teacher-student mechanism. When there is no intra-image tutor, the false label with lower accuracy leads to bias in the training target. The deviation affects the optimization of all model parameters in subsequent iteration, resulting in degraded detection performance. When there is no inter-image tutor, joint training on the synthetic target image and the labeled target image leads to degraded performance because it is easy to ignore the prior knowledge from the source domain image. This shows that when there is only one single tutor, the prior knowledge in the unlabeled target domain or the labeled source domain is not effectively exploited. The two-tutor model performs better than the no-tutor and one-tutor model, and our DSLN framework reaches accuracy values of 0.9814, 0.9652 and 0.9845, respectively. The above experimental results show that using inter-image tutor and intra-image tutor can make full use of the prior knowledge from the labeled source images and unlabeled target images. They all contribute to the accurate glaucoma detection to varying degrees, and the combination of the two can achieve best experimental result.

4.6 Computation time

To analyze the computational time complexity of our method, we compared the training and test time of different methods, and the results are shown in Table 11. To ensure the fairness of the comparison, we train each network on a single Nvidia GeForce Titan GPU for 100 epochs in the same experimental settings. It can be seen from Table 11 that our method takes 7 hours to train 100 epochs on the GPU, and it takes 0.15 seconds to test a fundus image with a resolution of 256×256. Compared with other methods, our method has the shortest calculation time.

The main reason is that we use the lightweight modified ResNet50 as backbone to extract features from the local optic disc area and the whole fundus image, respectively. ResNet50 introduces a residual block through skip connection to speed up the flow of information, and it is merged by multiple shallow networks to speed up network convergence, thereby shortening training time. Li et al. [20] successively adopted attention prediction subnet, pathological area location subnet and glaucoma classification subnet to realize glaucoma detection. However, training each subnet takes much time, so the overall training time of this method is longer than that of the proposed method. Chai et al. [23] used a multi-stream network to integrate non-image features and fundus image features for automatic glaucoma detection. This method uses a fully convolutional neural network to segment the peripapillary atrophy, optic disc and optic cup, but more calculation parameters will be generated as the number of network layers increases, which consumes computing resources and increases the training time.

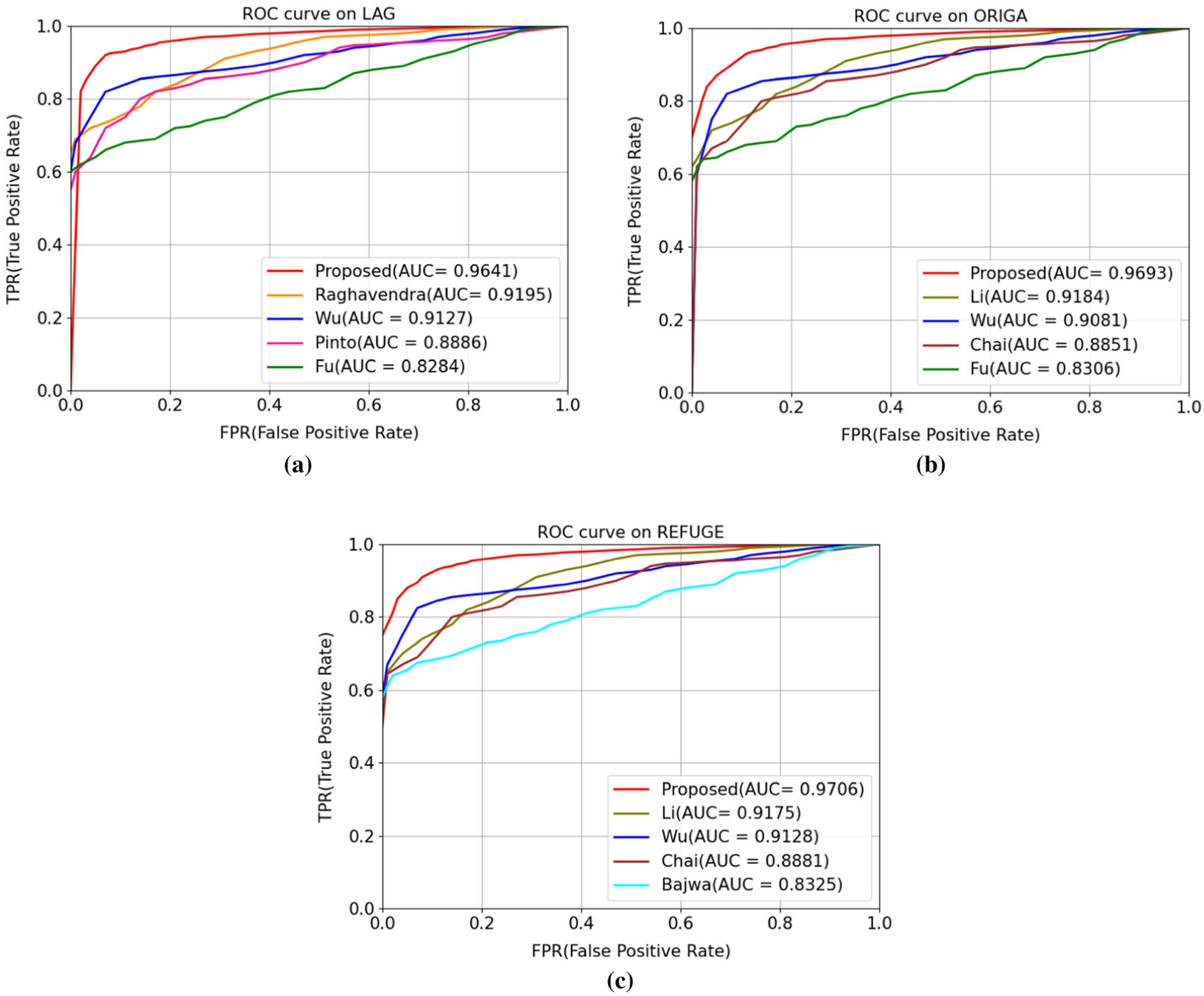

Fig. 12 ROC curves on three datasets for glaucoma screening **a** ROC curve on LAG; **b** ROC curve on ORIGA; **c** ROC curve on REFUGE

5 Discussion

We now analyze and discuss the performance of different backbone networks, the role of each branch in the backbone network and the integration way.

(1) *Backbone network analysis:* To evaluate the performance of different backbone networks, Table 12 compares five types of backbone networks [28, 29, 34, 46] in terms of parameter quantity, classification performance and test time cost. It can be seen from Table 12 that the AUC values of VGG16 [29] and VGG19 [29] are higher than ResNet [28], but more calculation parameters generate as the number of network layers gradually deepens, which requires more computing power and resources. ResNet50 [28] introduces skip connections to make the information of the previous residual block flow into the next residual block without hindrance, which helps to improve the information flow. Moreover, it can avoid the problem of vanishing gradient and degradation due to too deep network. Compared with other backbones, ResNet50 has fewer parameters, and the test time is much shorter than other backbones while guaranteeing better classification performance.

(2) *Branch analysis:* We analyze the performance of different branches of the backbone network on LAG and REFUGE, and the results are shown in Fig. 15. It is observed that the performance of the local optic disc branch is better than that of the global image level, mainly because most of the clinical symptoms of glaucoma are manifested near the optic disc area, which is consistent with the experience of ophthalmologists in diagnosing glaucoma. However, the experimental performance of fusing the two levels of local optic area and global fundus image is the best. It shows that the local optic disk and the

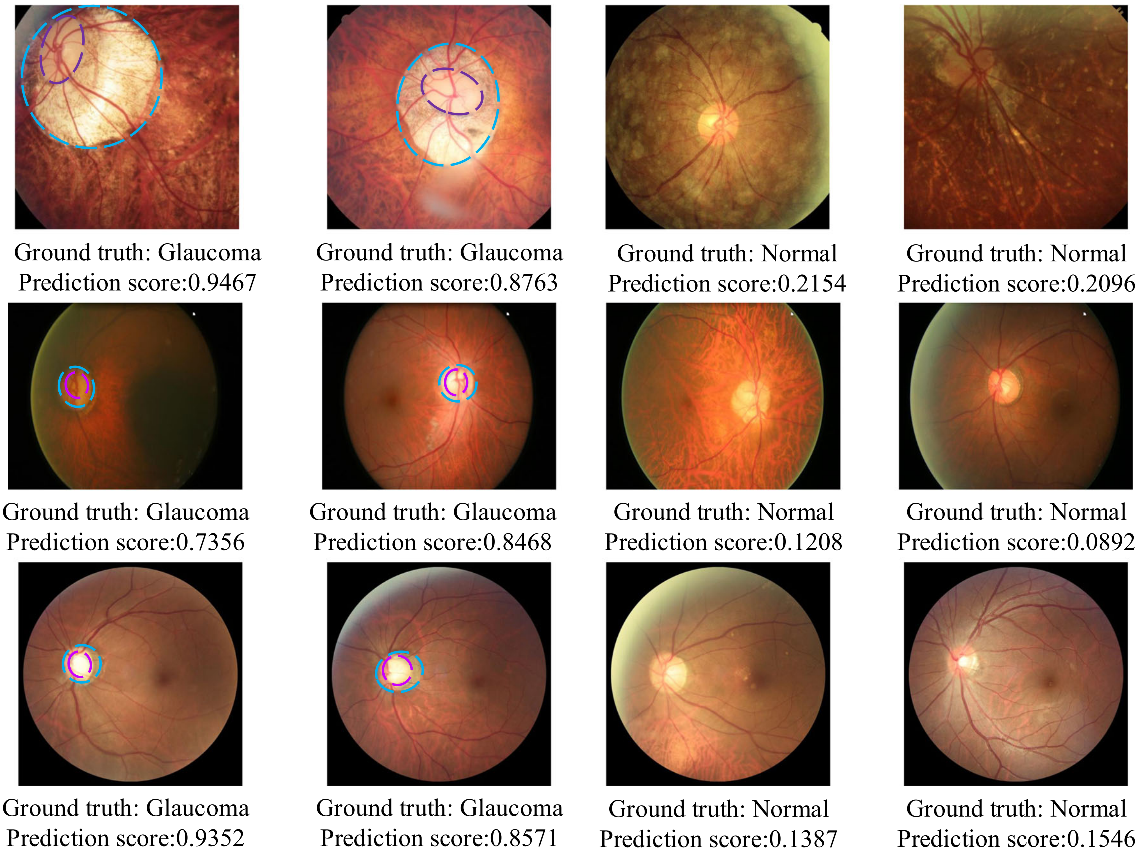

Fig. 13 Correct classification results of glaucoma and normal eyes on LAG, ORIGA and REFUGE. The purple and blue ellipses represent detected lesion areas of glaucoma

whole fundus image branch play a vital role in our backbone network, and the combination of the two helps to improve the performance of glaucoma detection.

(3) *Analysis of integration ways*: We tested the fusion ways of different branches on LAG and REFUGE, including maximum, minimum, average and multiplication operations. The comparison results are shown in Fig. 16. It is observed from Fig. 16 that no matter which fusion operation is used for the two branches, the performance of glaucoma detection can be effectively improved, and even the minimum operation is better than one single branch. This indicates that the fusion operation helps to improve the performance of glaucoma screening. Moreover, the experimental performance of averaging the two branches on LAG and REFUGE is best. Therefore, our backbone network uses averaging to integrate the outputs of the two branches.

There is great class imbalance between glaucoma and normal eyes in some datasets (e.g., REFUGE, ORIGA, and RIM-ONE-r1), the proposed DSLN uses the weighted sum of class-balanced binary cross-entropy loss and dice loss as the overall loss function to train the network, which can effectively alleviate the issue of data imbalance. It is observed from Tables 3 and 8 that the accuracy and AUC value of the proposed framework on RIM-ONE-r1 and ORIGA reached 0.9764/0.9693 and 0.9748/0.9760, respectively. Compared with the method presented by Elangovan et al. [43], the accuracy and AUC value are

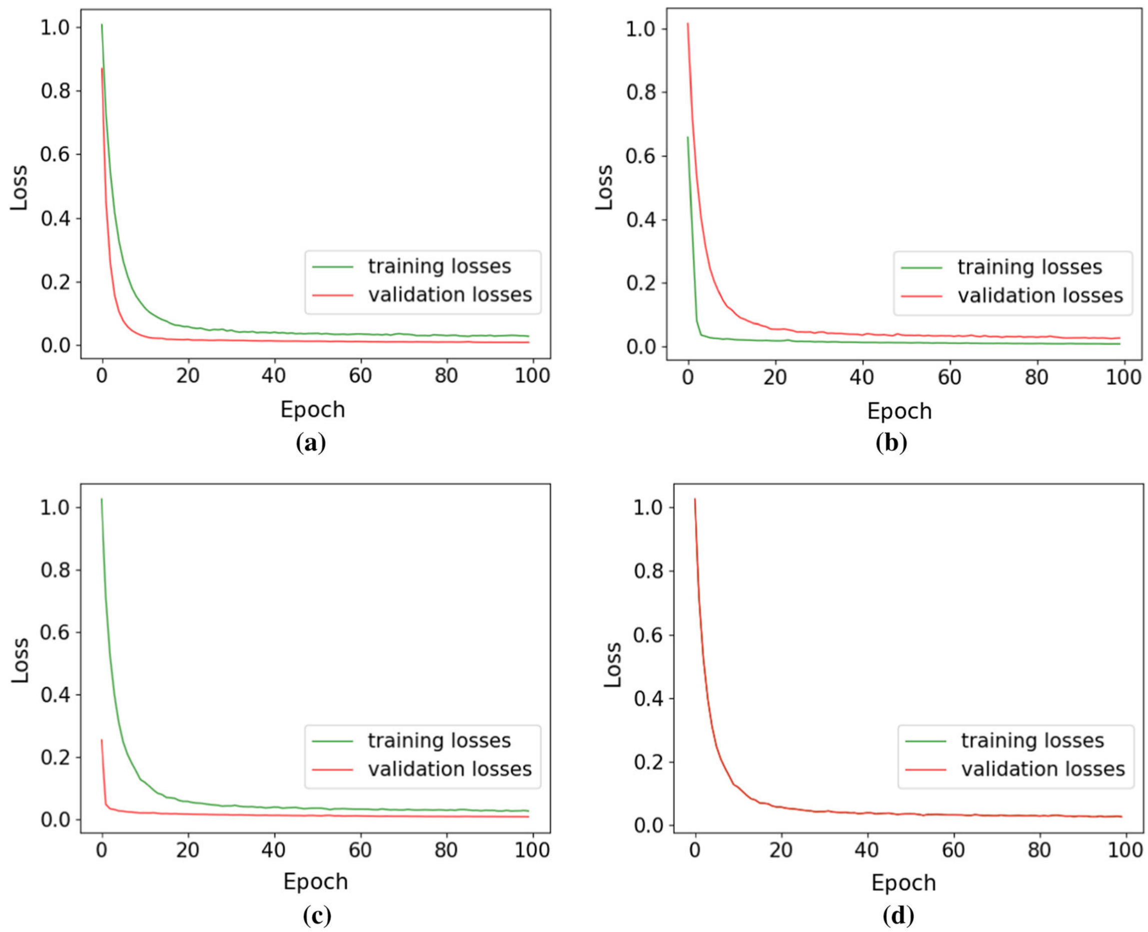

Fig. 14 Convergence curve on four datasets **a** loss curve on ACRIMA; **b** loss curve on DRISHTI-GS; **c** loss curve on LAG; **d** loss curve on REFUGE

Table 10 Ablation results on LAG, ACRIMA and DRISHTI-GS

Methods/evaluation criteria		LAG		ACMIRA		DRISHTI-GS	
		Acc	AUC	Acc	AUC	Acc	AUC
No-tutor	Baseline	0.9658	0.9671	0.9576	0.9645	0.9625	0.9547
	GAN baseline	0.9684	0.9715	0.9595	0.9613	0.9745	0.9634
One-tutor	w/o intra-image	0.9717	0.9748	0.9622	0.9686	0.9789	0.9638
	w/o inter-image	0.9763	0.9817	0.9634	0.9691	0.9752	0.9463
Two-tutor(ours)		**0.9814**	**0.9831**	**0.9652**	**0.9726**	**0.9845**	**0.9704**

The best results are shown in bold

improved by 11.57%/8.02% and 19.32%/4.42, respectively.

Although the proposed method can detect most glaucomatous optic neuropathy in different datasets, there are still false positive and false negative results. As shown in Fig. 17, the presence of other fundus diseases increases the difficulty of glaucoma detection, resulting in misclassified fundus images. It is found that high or pathological myopia is the main cause of false positive and false negative results. Thinning of the retinal nerve fiber layer (RNFL) and tilting of optic nerve head are common clinical manifestations of high myopia, which are highly like the pathological features of glaucoma, so it is easy to misclassify high myopia as glaucoma. In addition, the

Table 11 Computation time comparison of different methods

Methods	Training time	Test time
Wu et al. [5]	8.5 h	0.5 s
Pinto et al. [6]	9.2 h	0.41 s
Li et al. [20]	9 h	0.35 s
Chai et al. [23]	10 h	1.2 s
Elangovan et al. [43]	8 h	0.7 s
Ours	**7 h**	**0.15 s**

The best results are shown in bold

Table 12 Comparison of different network backbones

Model	Parameters	Acc	AUC	Time
ResNet50 [28]	**20 M**	**0.9781**	**0.9735**	**0.15 s**
VGG16 [29]	138 M	0.9486	0.9754	1.21 s
VGG19 [29]	144 M	0.9537	0.9787	0.79 s
Inceptionv4 [34]	25 M	0.9653	0.9548	0.86 s
Xception [46]	22 M	0.9568	0.9614	0.54 s

The best results are shown in bold

physiological large optic cup and congenital optic disc abnormalities could cause false positive results, while age-related macular degeneration and diabetic retinopathy are prone to false negative results. Therefore, how to accurately detect glaucoma images with coexisting other eye diseases will be the focus of our future research.

6 Conclusion

This paper proposes a dual-tutor student learning network (DSLN) to solve the issue of domain shift in fundus images from different race for accurate glaucoma detection. It can simultaneously use the rich information of abundant unlabeled data and limited labeled data. The inter-image tutor and intra-image tutor model learn the prior knowledge from the labeled source images and unlabeled target images, respectively, and meanwhile transfer the inter-image and intra-image knowledge to the student model for comprehensive optimization and integration. It can effectively combine the advantages of domain adaptation and semi-supervised learning. We conducted experimental on nine datasets (e.g., RIM-ONE-1, DRISHTI-GS, LAG and ORIGA) from three races and obtained good performance. Experimental results show that the proposed DSLN framework has good robustness and accuracy. It can not only overcome the adverse effect of domain shift and noise effectively, but also detect glaucoma in multi-ethnic fundus images accurately.

However, the coexistence of other fundus diseases may affect the accuracy of glaucoma detection. In the future, we will further explore how to reduce the false positive and false negatives when other fundus diseases exist for

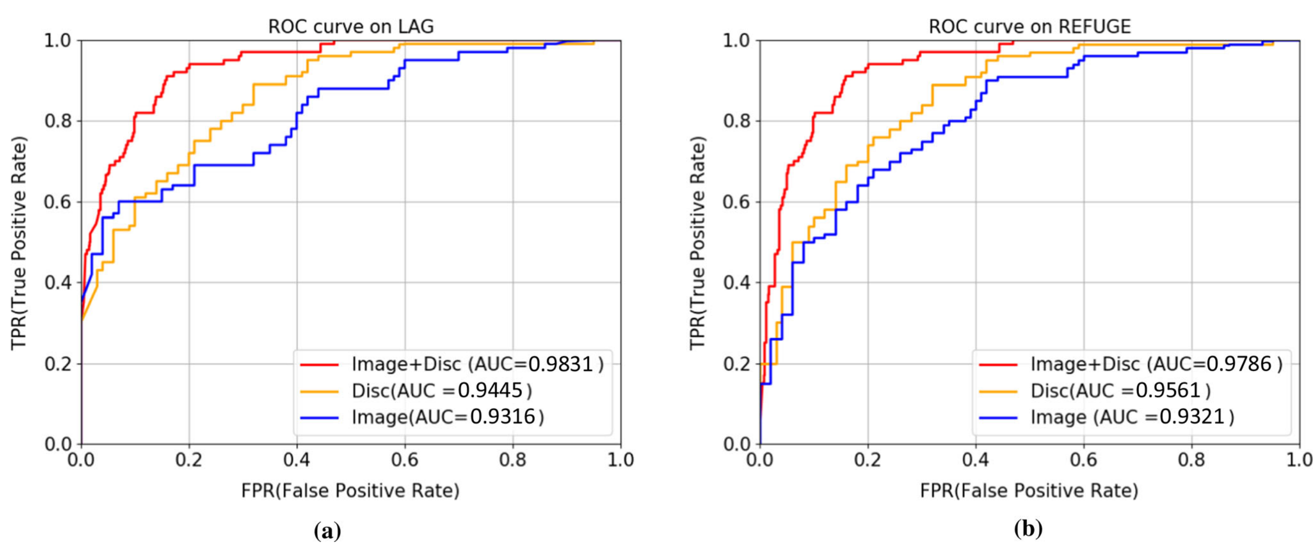

Fig. 15 The performances of different streams on the LAG and REFUGE datasets

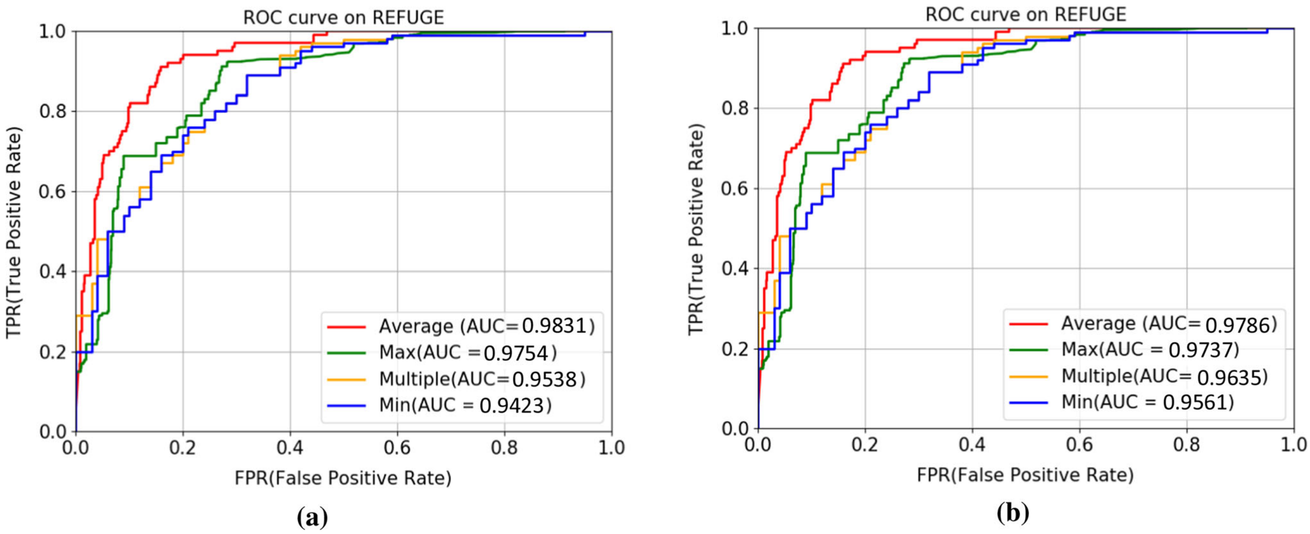

Fig. 16 The performance of various fusion operation on the LAG and REFUGE datasets

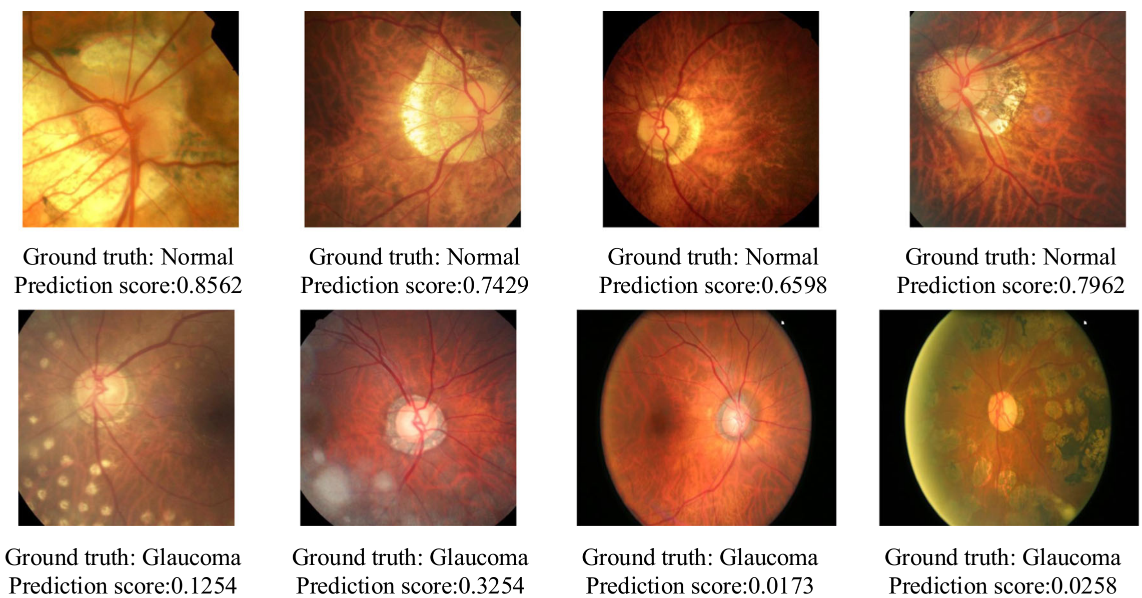

Fig. 17 Examples of misclassified glaucoma and healthy images

accurate glaucoma detection. Moreover, compared with 2D color fundus images, OCT images have discriminative features for improving the performance of glaucoma detection. Although 3D OCT images are expensive and difficult to obtain, it does not deny the value of 3D OCT as a glaucoma screening tool. We will carry out further research on glaucoma detection based on 3D OCT images.

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Author contributions Yanfei Guo contributed to conceptualization, methodology, software, writing the original draft. Yanjun Peng helped in data curation, supervision. Jindong Sun contributed to investigation, formal analysis, software. Dapeng Li contributed to visualization, writing—review and editing. Bin Zhang contributed to resources, project administration.

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Availability of data and materials Data related to the current study are available from the corresponding author on reasonable request.

Declarations

Conflicts of interest The authors declare that they have no conflicts of interest.

Code availability Some of the codes generated or used during the study are available from the corresponding author by request.

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